

	Work Instruction	Doc. Revision	00
	AXI V810 XY-Axes Quantic Home and Limit Sensors Upgrade Alignment Procedure (AKD, S200, Panasonic)	Effective Date	15 Aug 2024

This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL			
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 AUG 2024	First Release	Yeap Leong Wei

	Work Instruction	Doc. Revision	00
	AXI V810 XY-Axes Quantic Home and Limit Sensors Upgrade Alignment Procedure (AKD, S200, Panasonic)	Effective Date	15 Aug 2024

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Scope

This procedure is designed to provide the guidelines for alignment of home and limit sensors that pairing with Quantic Scale.

Tool List

No	PN	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Camera Tool	1	PC	

Table 1: Tool List

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Home and Limit Sensors Alignment **for AKD and S200 Drive**

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X-Axis Home and Limit Sensor Alignment for AKD and S200 Drive Only

Note: for Panasonic Drive, proceed to [link](#)

1. **Launch the v810:** Find the v810 shortcut icon at the desktop window and launch the program.



Figure 1: v810 shortcut

2. **At V810 GUI, Stop the RMS:** Open V810. Navigate to ‘Services > Subsystem Status & Control > Connected Processors’ and click “Stop RMS”. Close the V810 upon RMS have been stopped.

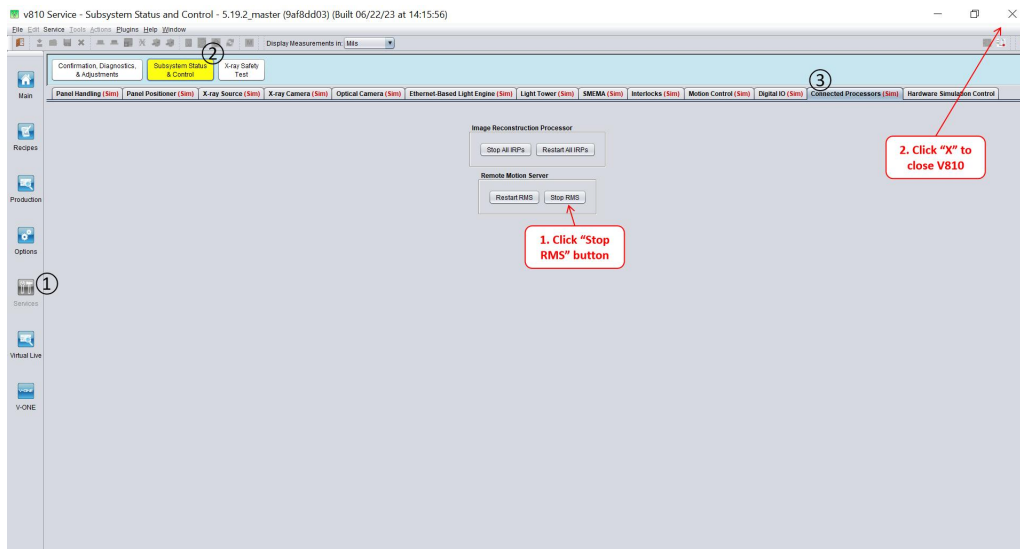


Figure 2: Stop RMS (Sample Image)

3. **Opening ‘hardware.config’ text file:** Navigate to ‘C:\Program Files\AXI System\v810\5.0*Config’. Open ‘hardware.config’ and search for “active_”. Refer to Kollmorgen model needed to be configured.
Note: *Config is dependent on machine model. Example S2EXconfig, XLTconfig or S3config.

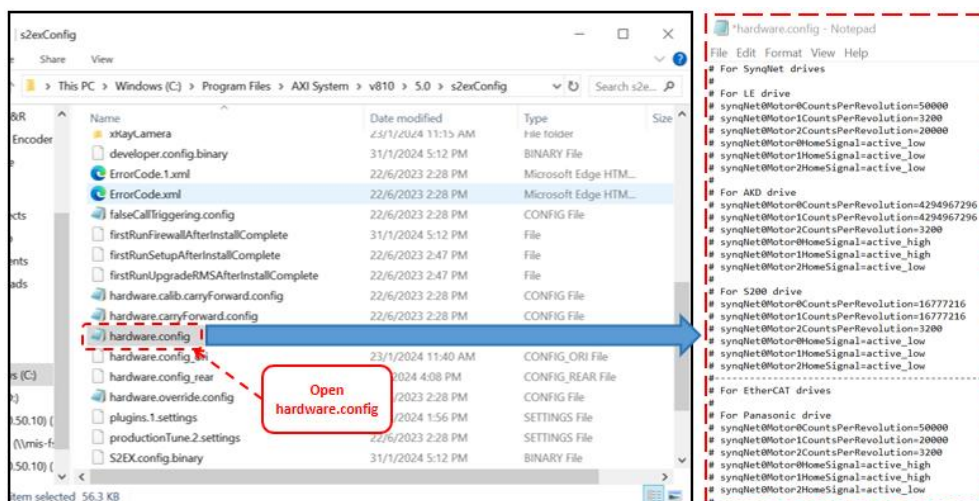


Figure 3: Open Hardware.config

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4. **Home Signal Configuration:** Configure the X-Axis home signals by inverting the command.

Note: Dependent on Quantic scale axis being replaced. Do ensure only the specific axis is being

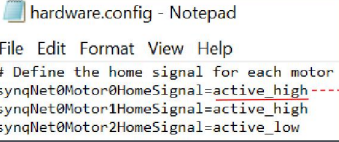
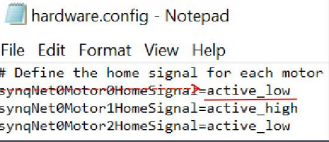
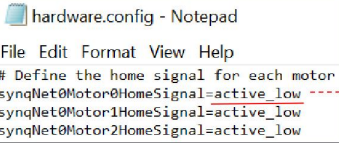
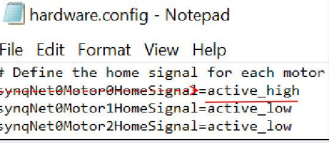
Kollmorgen Drive Model	Before Signal Configuration for X-Axis in "hardware.config"	After Signal Configuration for X-Axis in "hardware.config"
AKD	 <pre> File Edit Format View Help # Define the home signal for each motor synqNet0Motor0HomeSignal=active_high synqNet0Motor1HomeSignal=active_high synqNet0Motor2HomeSignal=active_low </pre>	 <pre> File Edit Format View Help # Define the home signal for each motor synqNet0Motor0HomeSignal=active_low synqNet0Motor1HomeSignal=active_high synqNet0Motor2HomeSignal=active_low </pre>
S200	 <pre> File Edit Format View Help # Define the home signal for each motor synqNet0Motor0HomeSignal=active_low synqNet0Motor1HomeSignal=active_low synqNet0Motor2HomeSignal=active_low </pre>	 <pre> File Edit Format View Help # Define the home signal for each motor synqNet0Motor0HomeSignal=active_high synqNet0Motor1HomeSignal=active_low synqNet0Motor2HomeSignal=active_low </pre>

Figure 4: X-Axis Home Signal Configuration (Before and After)

Configuration Change					
Drive	RGH22 Encoder		Quantic Encoder		
	AKD	S200	AKD	S200	Path
synQnet0Motor0Home Signal (X-Axis)	active_high	active_low	active_low	active_high	hardware.config
synqNet0Motor1HomeSignal (Y-Axis)	active_high	active_low	active_low	active_high	hardware.config

Table 2: X&Y-Axis Home Signal Configuration (Summary)

5. **Launch the V810 GUI and Restart RMS:** Open V810. Navigate to ‘Services > Subsystem Status & Control > Connected Processors’ and click “Restart RMS”.

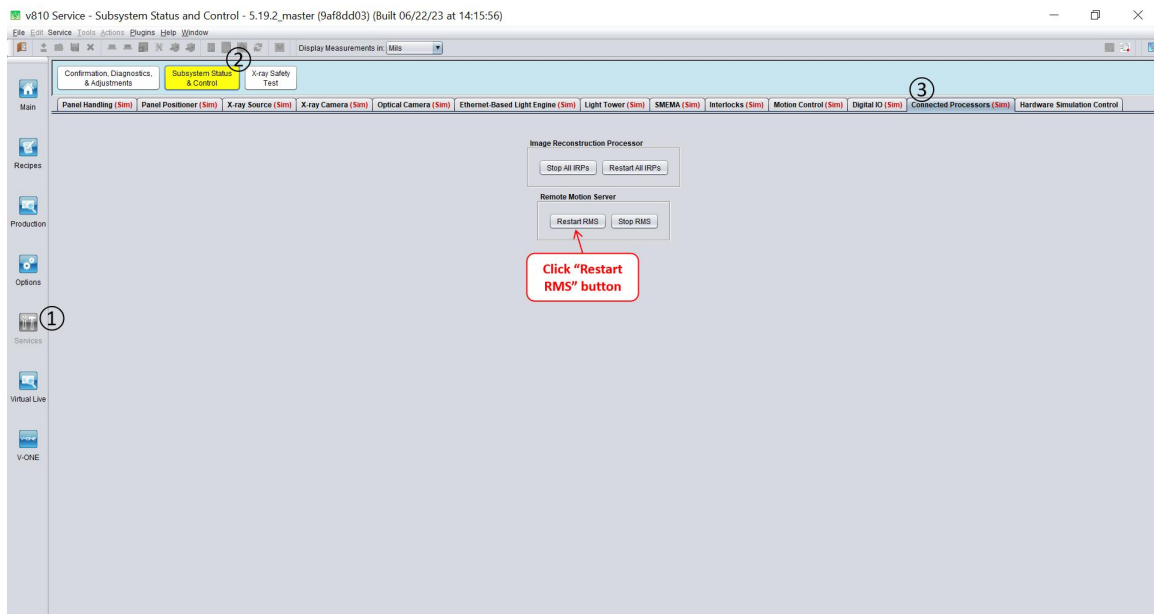


Figure 5: Restart RMS (Sample Image)

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6. **Digital IO Initialization:** Navigate to ‘Service > Subsystem Status & Control > Digital IO tab’. Click “Initialize” and wait for the subsystem initialization to complete.

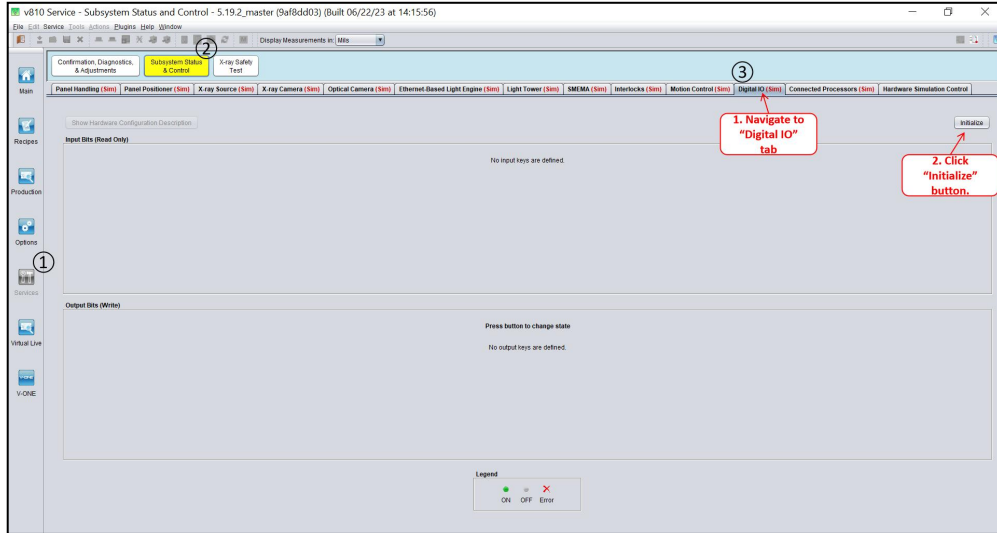


Figure 6: DIO Initialization (Sample Image)

7. **Panel Positioner Initialization:** Upon completion of DIO initialize, navigate to ‘Panel Positioner’ tab and click “Initialize”.

#IMPORTANT This step must be successfully executed before proceed to sensors adjustment.

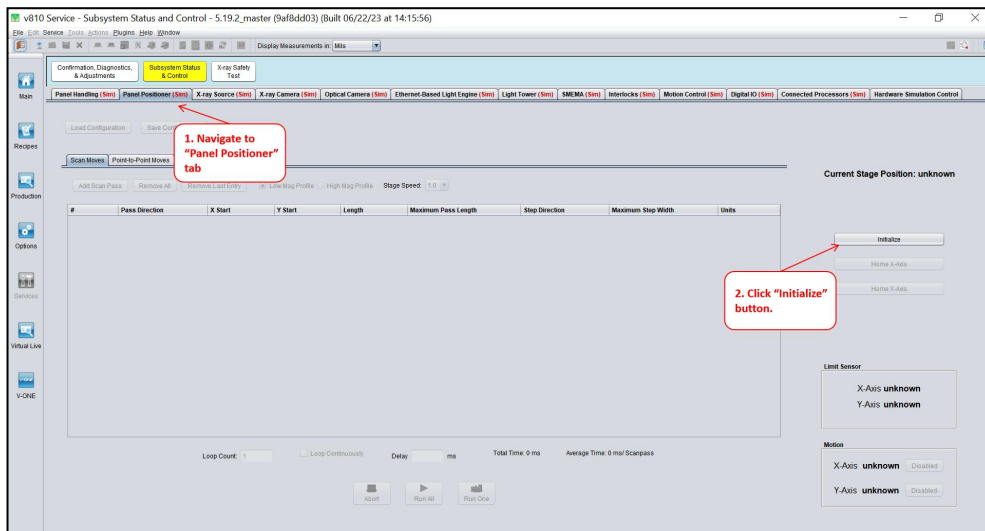


Figure 7: Initialize “Panel Positioner” (Sample Image)

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8. **Verify the “Current Stage Position” for X-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the X stage to the hard stop (maximum left) position and check if X-axis is within 3.08mm to 3.28mm. If not within the range, proceed to Step 9. Else, proceed to Step 13.

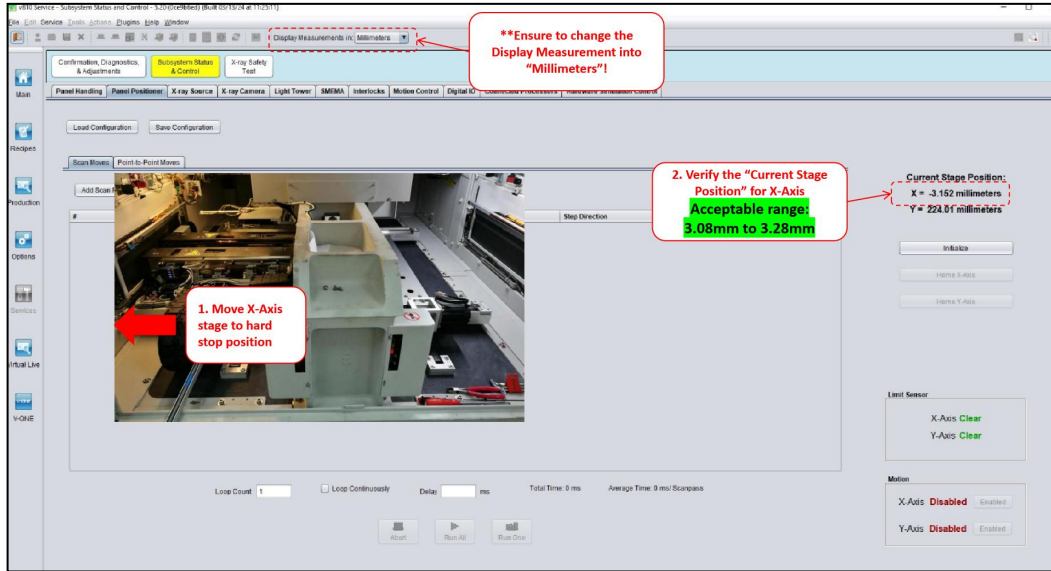


Figure 8: Current Stage Position for X-Axis within range (Sample Image)

9. **Loosen Spring Clamp:** Loosen all the screws for the spring clamp on X-axis.

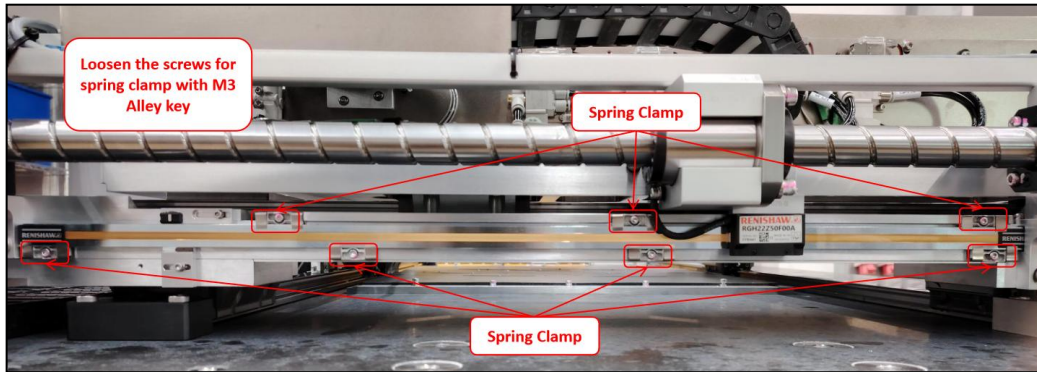


Figure 9: Quantic Scale Spring Clamps (Sample Image)

10. **Glass Scale Positioning Adjustment:** Gently move the glass scale following the direction indicated in the table below until the x-axis current position reading in V810 fall in the range between 3.08mm to 3.28mm.

Current stage position after initialize the Panel Positioner	Direction of the home sensor need to be adjusted
Less than -3.08mm	Towards the right outer barrier direction
More than -3.28mm	Towards the left outer barrier direction

Table 3: X-Axis Glass Scale Positioning Adjustment Guidelines

	Work Instruction	Doc. Revision	00
	AXI V810 XY-Axes Quantic Home and Limit Sensors Upgrade Alignment Procedure (AKD, S200, Panasonic)	Effective Date	15 Aug 2024

11. **Tighten the Spring Clamps:** Upon completion of X-Axis glass scale alignment, tighten all screw for the spring clamps.

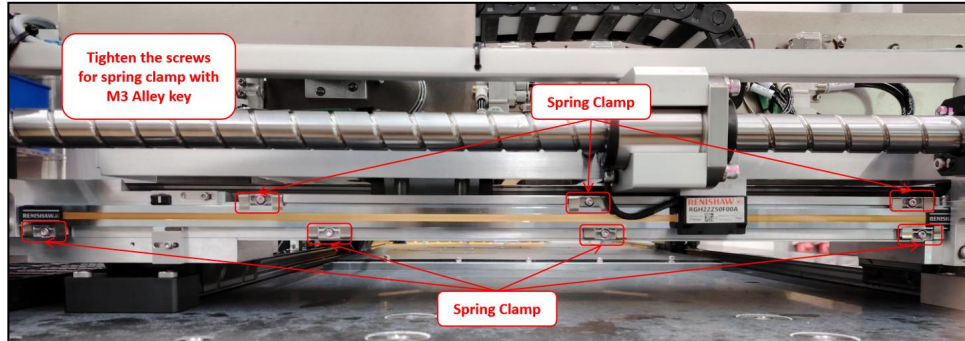


Figure 10: X-Axis glass scale Spring clamp (Sample Image)

12. **Home Sensor Positioning Offset:** Loosen the screw on the X-axis home sensor. Position the left side of the home sensor on the engraved line marking at the top side of the glass scale. Tighten back the screw on the X-axis home sensor when done.

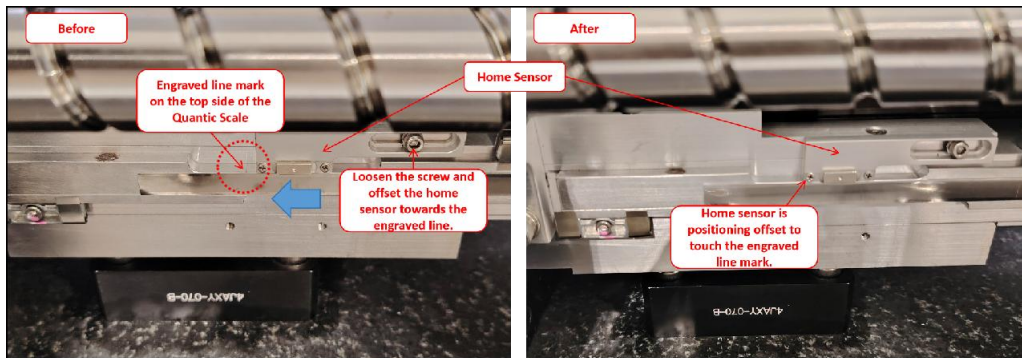


Figure 11: X-Axis Home Sensor Offset

13. **X-Axis Positioning (Away from Hard Stop):** Move the X-axis to the middle of the stage.

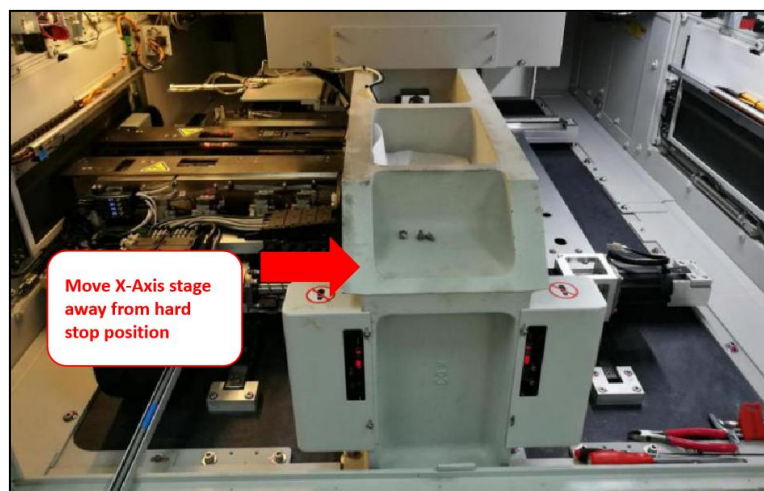


Figure 12: Move X-Axis Stage Away From Hard Stop

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14. **X-Axis Homing:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for X-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home X-Axis”.

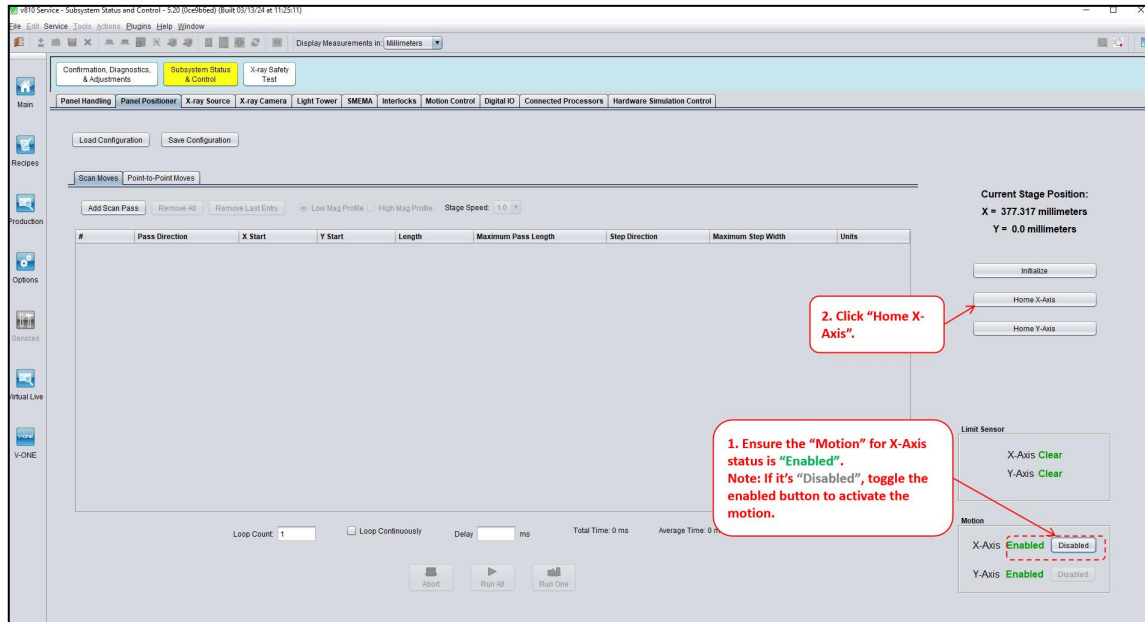


Figure 13: X-Axis Homing

15. **X-Axis “Current Stage Position” Verification (Hard Stop Position):** Move the X-Axis stage to hard stop position, double confirm the current stage position is within the range of 3.08mm to 3.28mm. If current position of X-Axis is out of range from 3.08mm to 3.28mm, move the x-stage to center and perform x-axis homing. Then, route back to Step 9. Else, move the stage to center position and proceed to the next step.

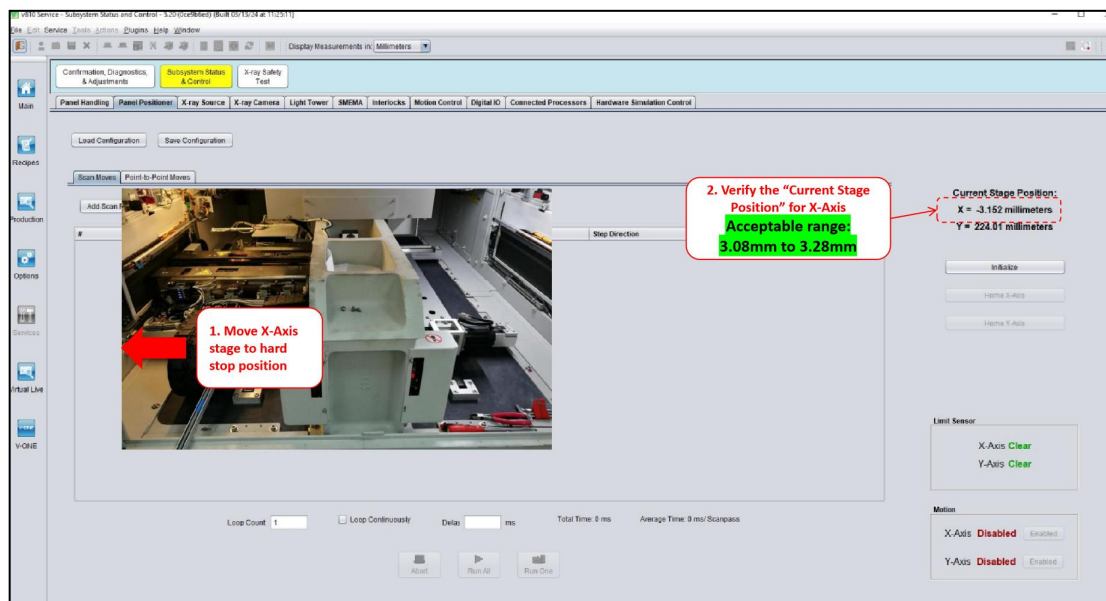


Figure 14: Verification of Current Stage Position for X-Axis

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16. **Homing again:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for X-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home X-Axis”

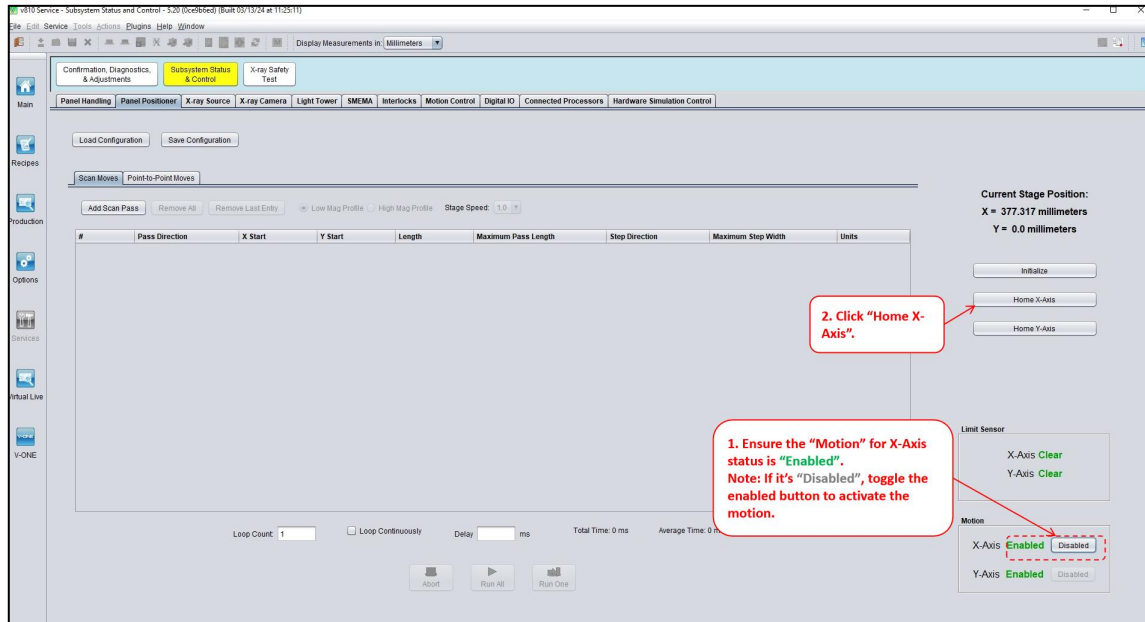


Figure 15: Perform Homing

17. **Move the Stage towards Hard Stop direction until it hit -0.5mm:** Move the stage towards the hard stop direction and ensure the current stage position is stop at -0.5mm.

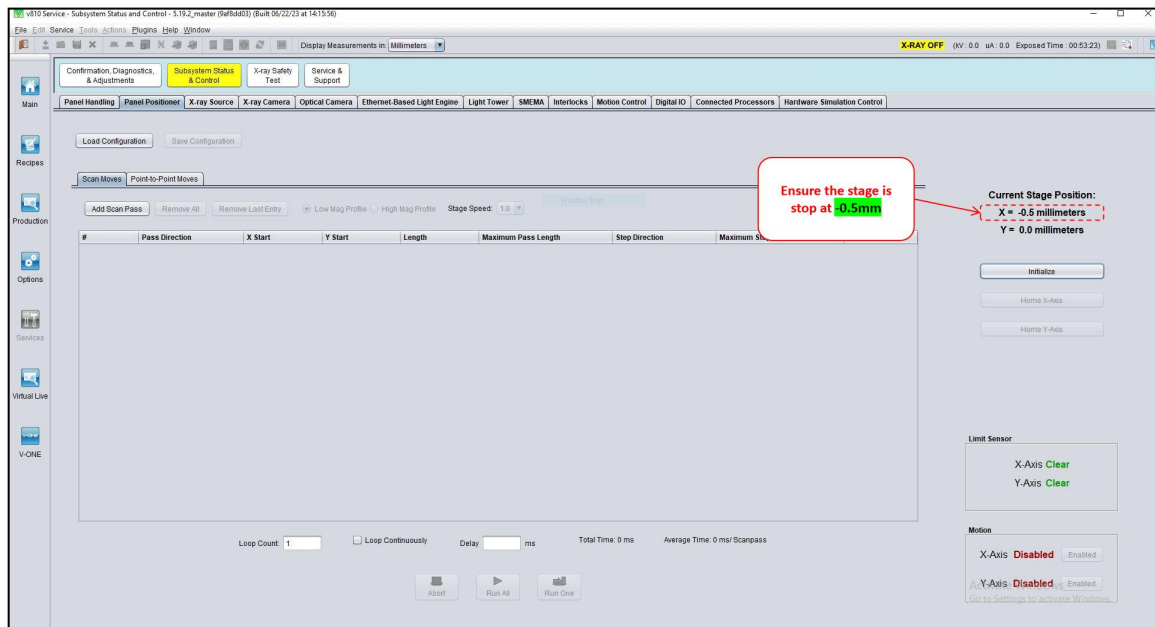


Figure 16: Move X-Axis Stage to Current Stage Position -0.5mm (Sample Image)

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18. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

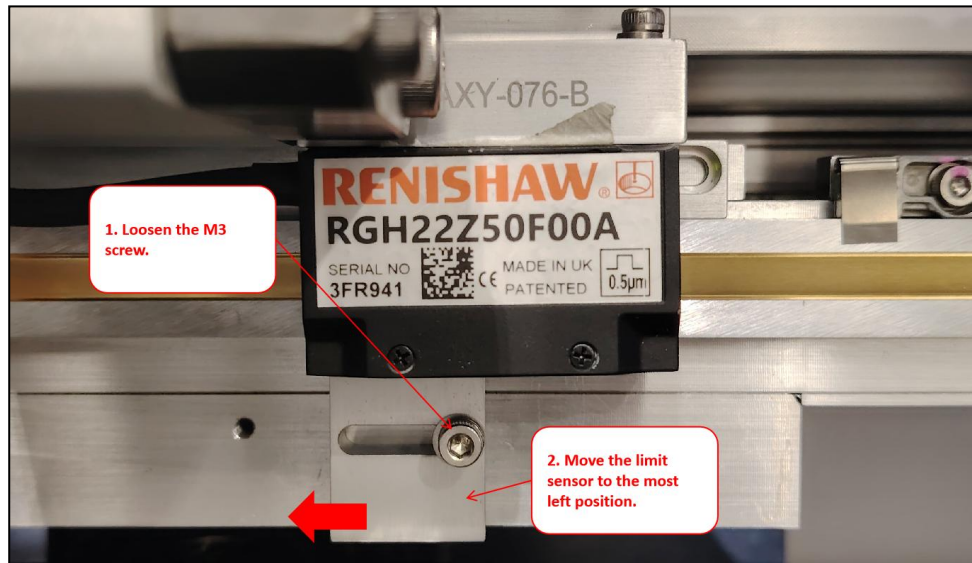


Figure 17: Move Limit Sensor to the Most Left Position

19. **Adjust the limit sensor to right hand side until the signal is “Blocked”:** Slowly move the limit sensor towards the right hand side position until the signal for limit sensor is blocked.

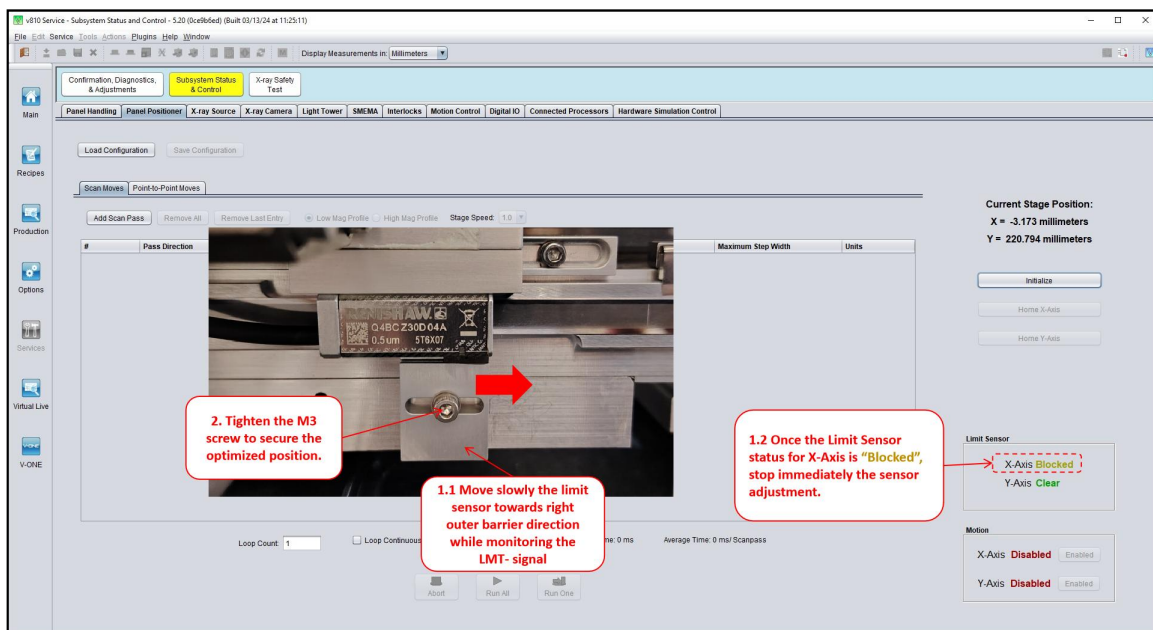


Figure 18: Positioning Limit Sensor to Triggered State

20. **Final Round Verification:** Lastly, manually move the X-stage to the center position, perform homing and ensure the signal for limit sensor is “Clear”. Then, manually move the stage to hard stop position and ensure the signal for limit sensor is “Blocked”.

21. **Validation of Machine Performance After Adjustment and Confirmation:** Machine performance validation is required before release back to customer. Refer to “[Validation of Machine Performance After Alignment for Sensors](#)” procedure in last page.

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Y Axis Home and Limit Sensor Alignment for AKD and S200 Drive Only

Note: for Panasonic Drive, proceed to [link](#)

22. **Launch the V810:** Find the v810 shortcut icon at the desktop window and launch the program.



Figure 19: v810 shortcut

23. **At V810 GUI, Stop the RMS:** Open V810. Navigate to ‘Services > Subsystem Status & Control > Connected Processors’ and click “Stop RMS”. Close the V810 upon RMS have been stopped.

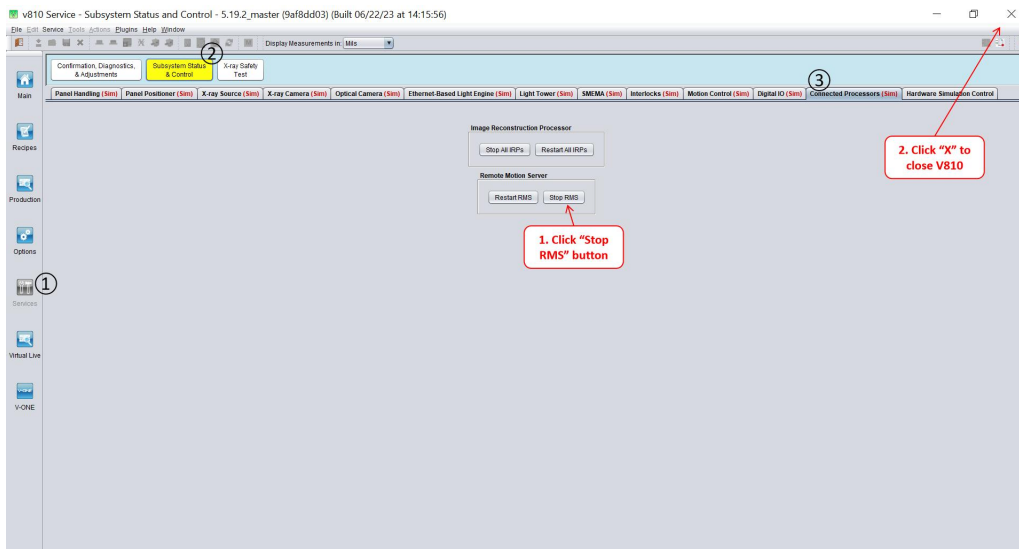


Figure 20: Stop RMS

24. **Opening ‘hardware.config’ text file:** Navigate to ‘C:\Program Files\AXI System\v810\5.0*Config’. Open ‘hardware.config’ and search for “active_”. Refer to Kollmorgen model needed to be configured. **Note:** *Config is dependent on machine model. Example S2EXconfig, XLTconfig or S3config.

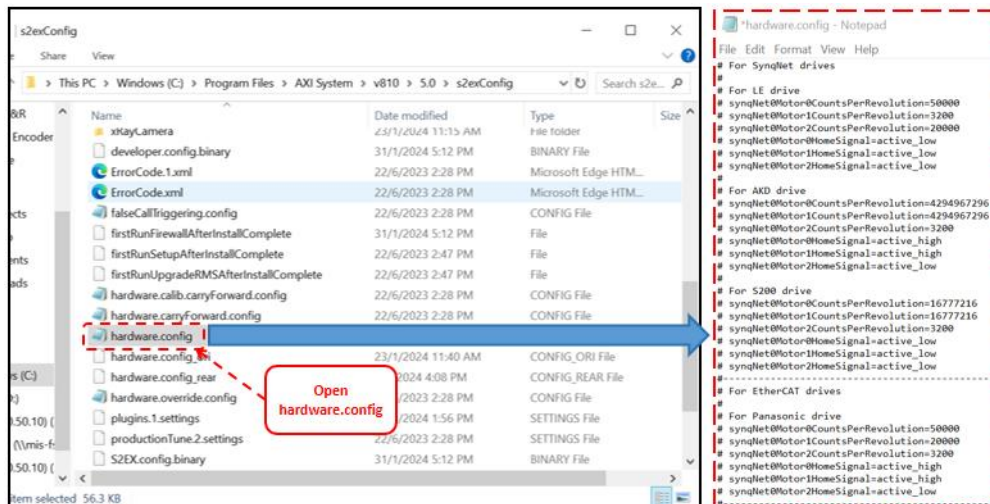


Figure 21: Open Hardware.config

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25. **Home Signal Configuration:** Configure the X-Axis home signals by inverting the command.

Note: Dependent on Quantic scale axis being replaced. Do ensure only the specific axis is being

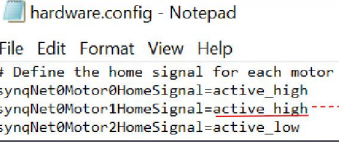
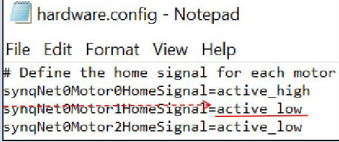
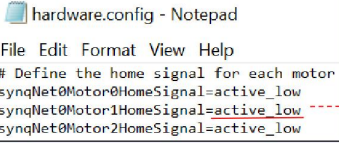
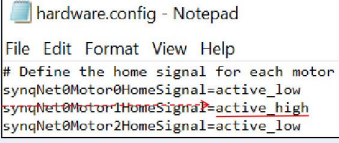
Kollmorgen Drive Model	Before Signal Configuration for Y-Axis in "hardware.config"	After Signal Configuration for Y-Axis in "hardware.config"
AKD		
S200		

Figure 22: Y-Axis Home Signal Configuration (Before and After)

Configuration Change					
Drive	RGH22 Encoder		Quantic Encoder		
	AKD	S200	AKD	S200	Path
synQnet0Motor0Home Signal (X-Axis)	active_high	active_low	active_low	active_high	hardware.config
synqNet0Motor1HomeSignal (Y-Axis)	active_high	active_low	active_low	active_high	hardware.config

Table 4: X&Y-Axis Home Signal Configuration (Summary)

26. **Launch the V810 GUI and Restart RMS:** Open V810. Navigate to ‘Services > Subsystem Status & Control > Connected Processors’ and click “Restart RMS”.

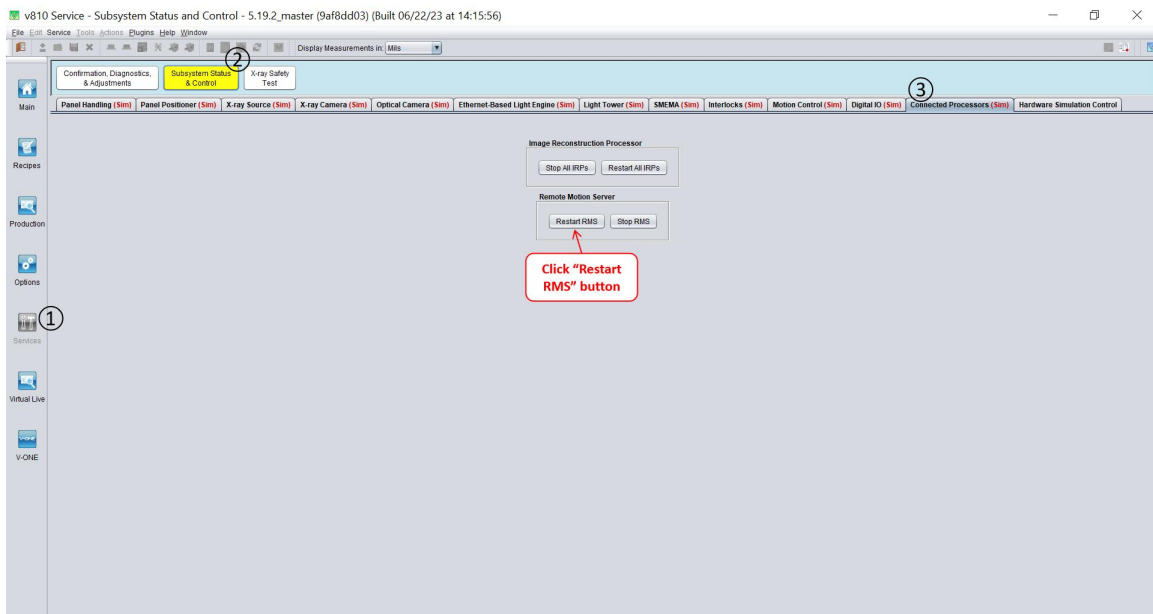


Figure 23: Restart RMS (Sample Image)

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27. **Digital IO Initialization:** Navigate to ‘Service > Subsystem Status & Control > Digital IO tab’. Click “Initialize” and wait for the subsystem initialization to complete.

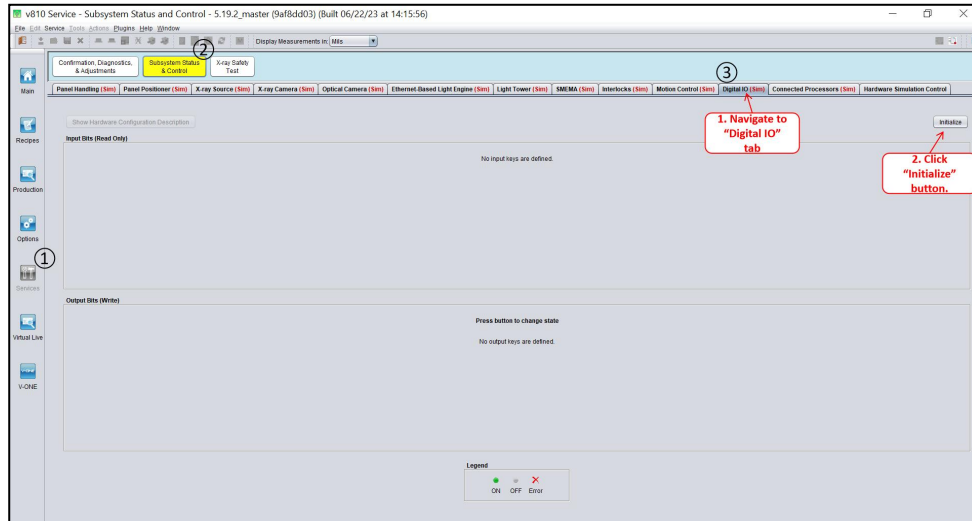


Figure 24: DIO Initialization (Sample Image)

28. **Panel Positioner Initialization:** Upon completion of DIO initialize, navigate to ‘Panel Positioner’ tab and click “Initialize”.

#IMPORTANT This step must be successfully executed before proceed to sensors adjustment.

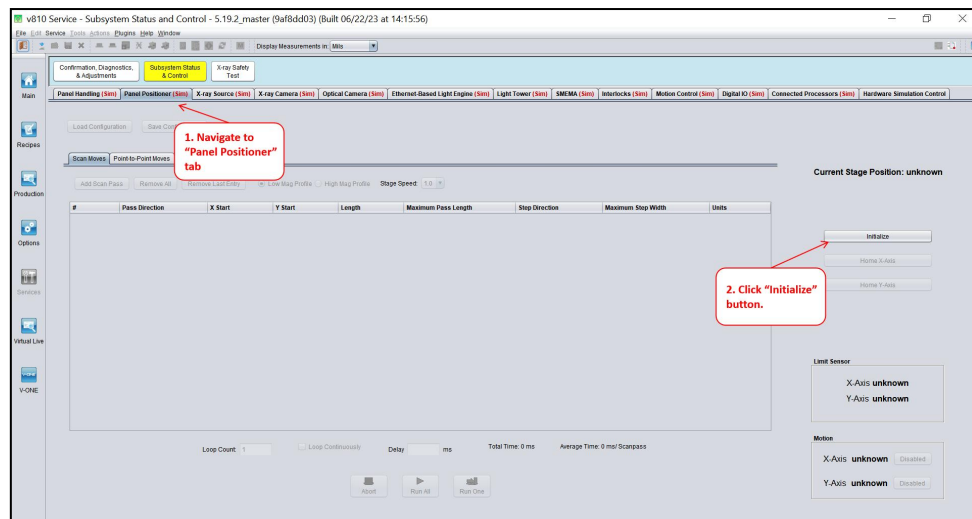


Figure 25: Initialize “Panel Positioner” (Sample Image)

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29. **Verify the “Current Stage Position” for Y-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the Y stage to the hard stop position and verify the home sensor reading is within 3.08mm to 3.28mm. If not within the range, proceed to Step 30. Else, proceed to Step 34.

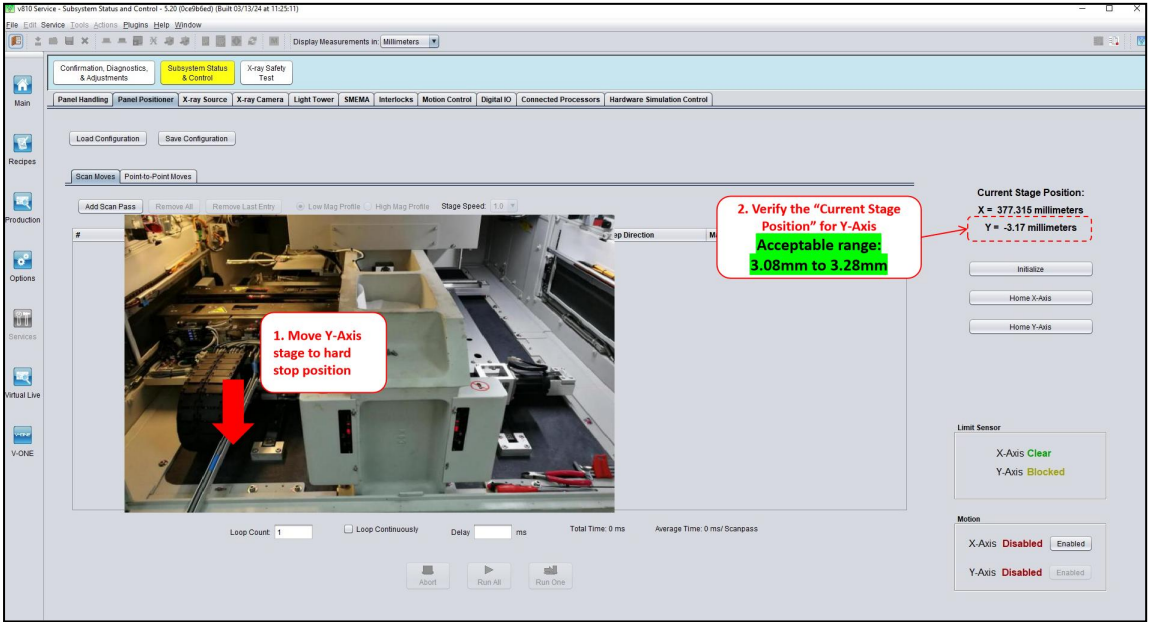


Figure 26: Current Stage Position for Y-Axis within range (Sample Image)

30. **Loosen the Screws for Spring Clamp and Limit Sensor:** Loosen all the screws for the spring clamp and limit sensor using precision tool. Be cautious not to damage the Quantic scale.

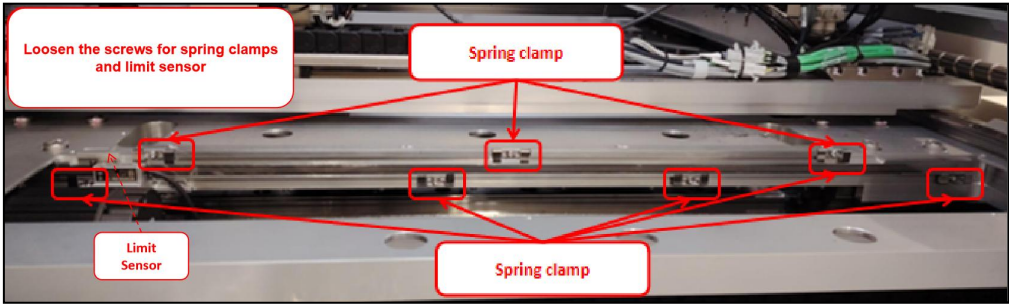


Figure 27: Quantic Scale Spring Clampers

31. **Quantic Scale Positioning Adjustment:** Gently move the Quantic scale as below table. While moving the Quantic scale, check the “Current Stage Position” in V810 and ensure the reading fall in the range of 3.08mm to 3.28mm.

Current stage position after initialize the Panel Positioner	Direction of the home sensor need to be adjusted
Less than -3.08mm	Towards the front access door
More than -3.28mm	Towards the rear access door

Table 5: Y-Axis Quantic Scale Positioning Adjustment Guidelines

	Work Instruction	Doc. Revision	00
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32. **Tighten the Screws for Spring Clamps and Limit Sensor:** Upon completion of Y-Axis home position alignment, tighten all screw for the spring clamps and limit sensor.

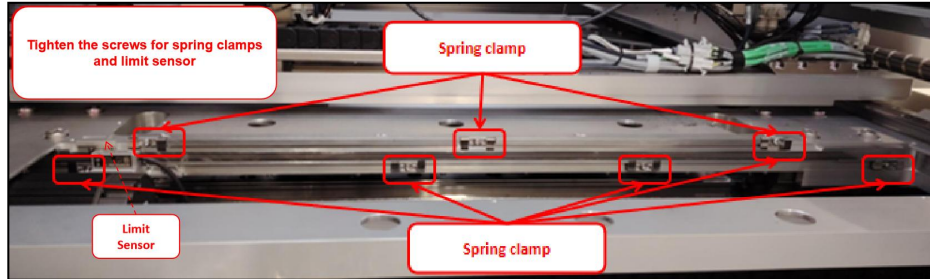


Figure 28: Y-Axis Quantic Scale Spring clamp

33. **Home Sensor Positioning Offset:** Move the position of the home sensor to cross the line of Quantic scale and stop at the right edge of the screw for the magnet sensor and tighten the screw.

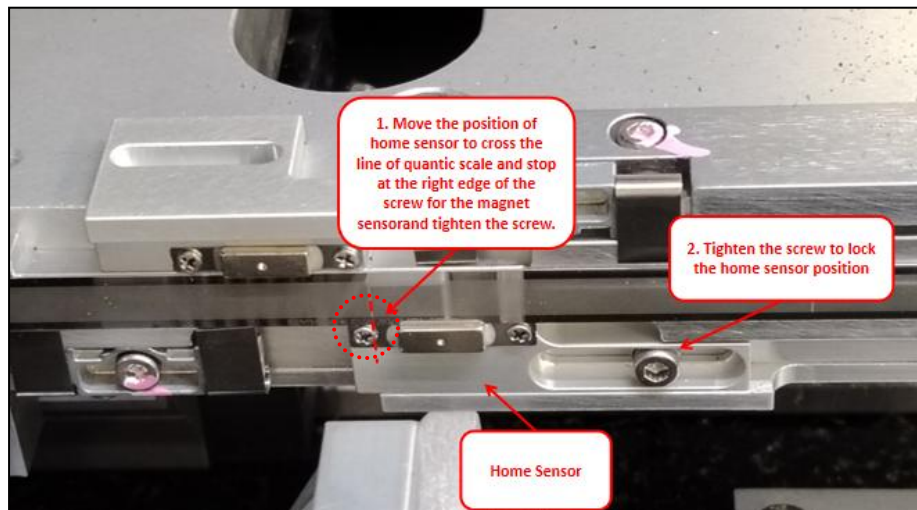


Figure 29: Y-Axis Home Sensor Offset

34. **Y-Axis Positioning (Away from Hard Stop):** Gently move the Y-Axis to the middle of the stage.

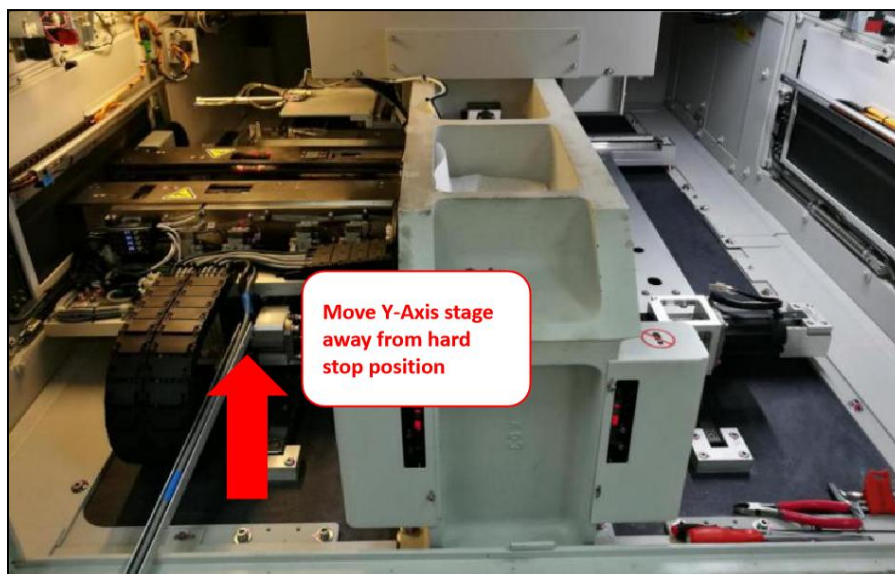


Figure 30: Move Y-Axis Stage Away From Hard Stop

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35. **Y-Axis Homing:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for Y-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home Y-Axis”

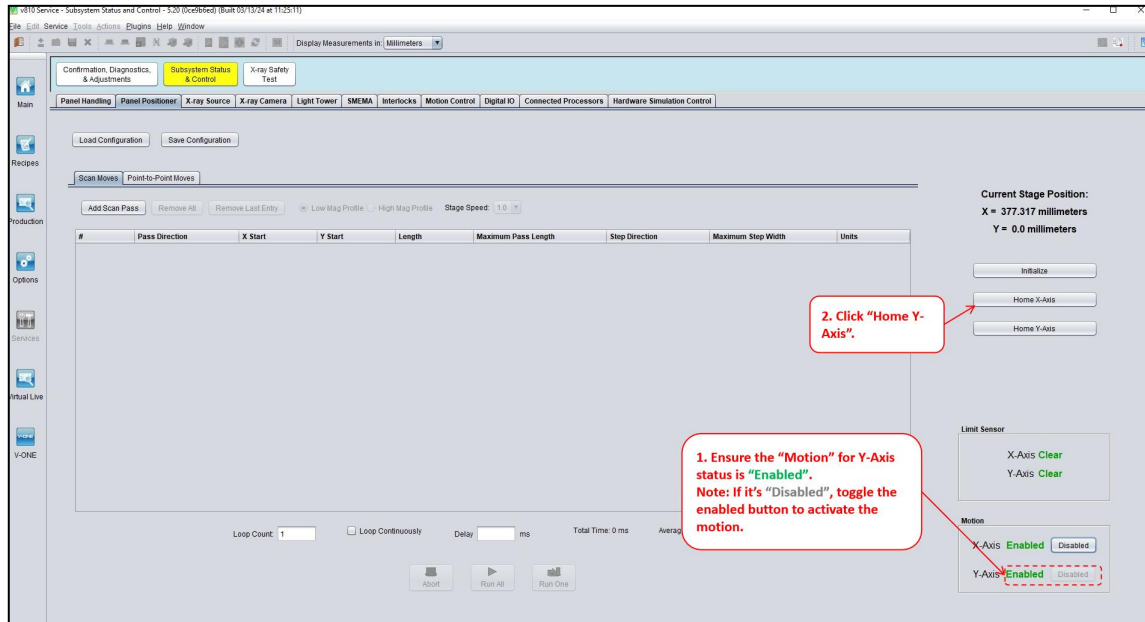


Figure 31: Y-Axis Homing

36. **Y-Axis “Current Stage Position” Verification (Hard Stop Position):** Move the Y-Axis stage to hard stop position, double confirm the current stage position is within the range of 3.08mm to 3.28mm. If current position of Y-Axis is out of range from 3.08mm to 3.28mm, route back to Step 30. Else, move the stage to center position and proceed to next step.

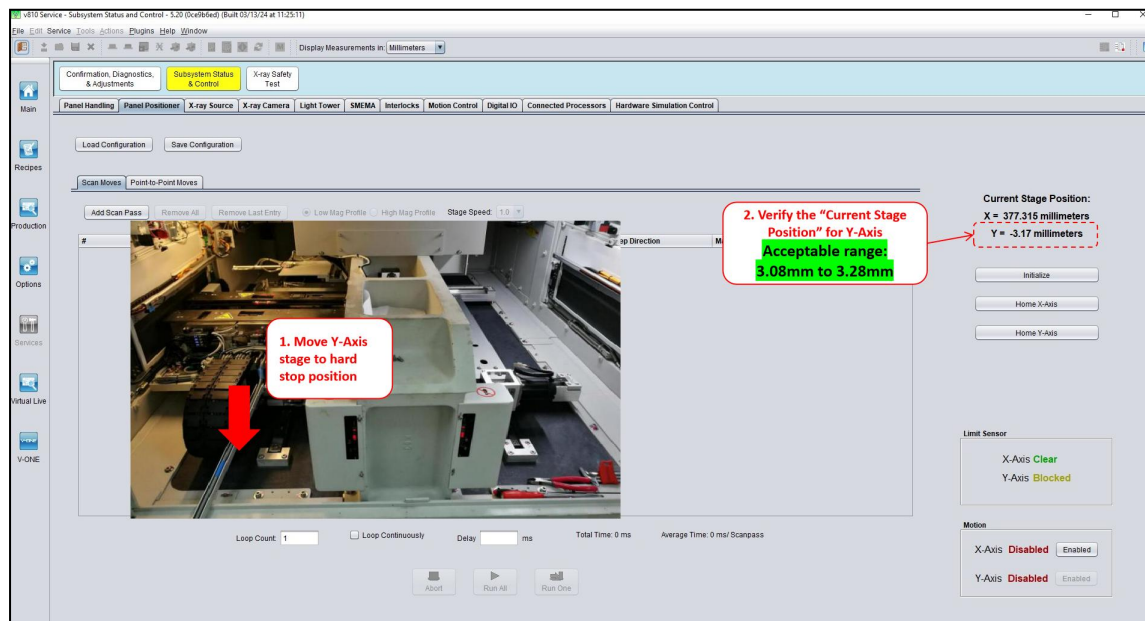


Figure 32: Verification of Current Stage Position for Y-Axis

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37. **Homing Again:** Navigate to ‘Panel Positioner’ tab, ensure the “Motion” for Y-Axis is enabled. If status is “Disabled”, toggle the enabled button to activate the motion. Then, perform homing by click “Home Y-Axis”

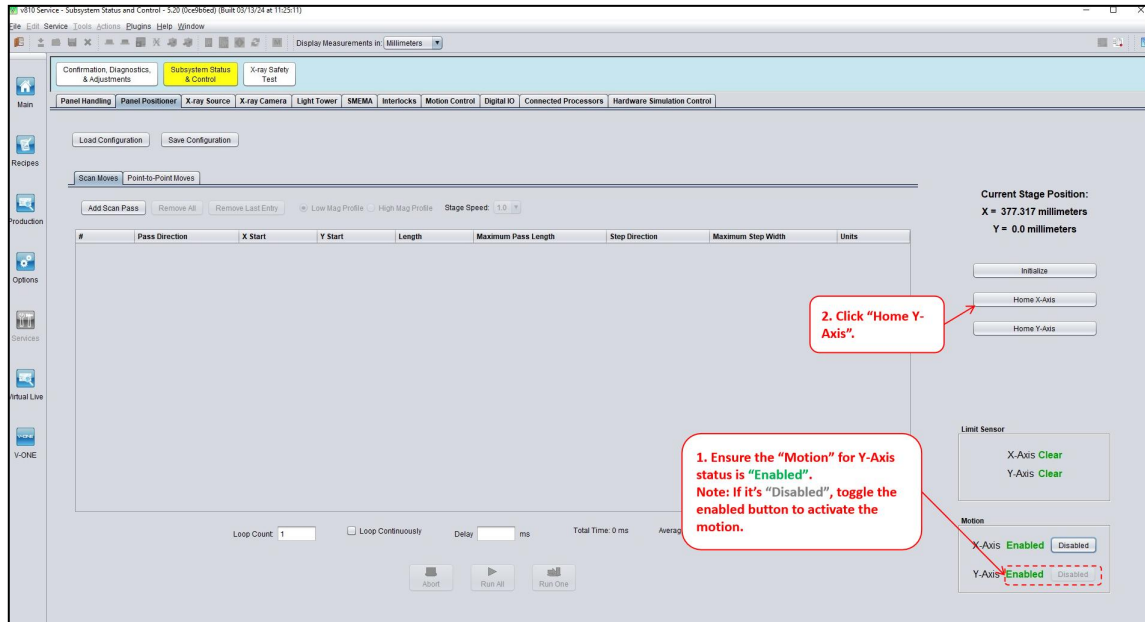


Figure 33: Y-Axis Homing

38. **Move the Stage towards Hard Stop direction until it hit -0.5mm:** Move the stage towards the hard stop direction and ensure the current stage position is stop at -0.5mm.

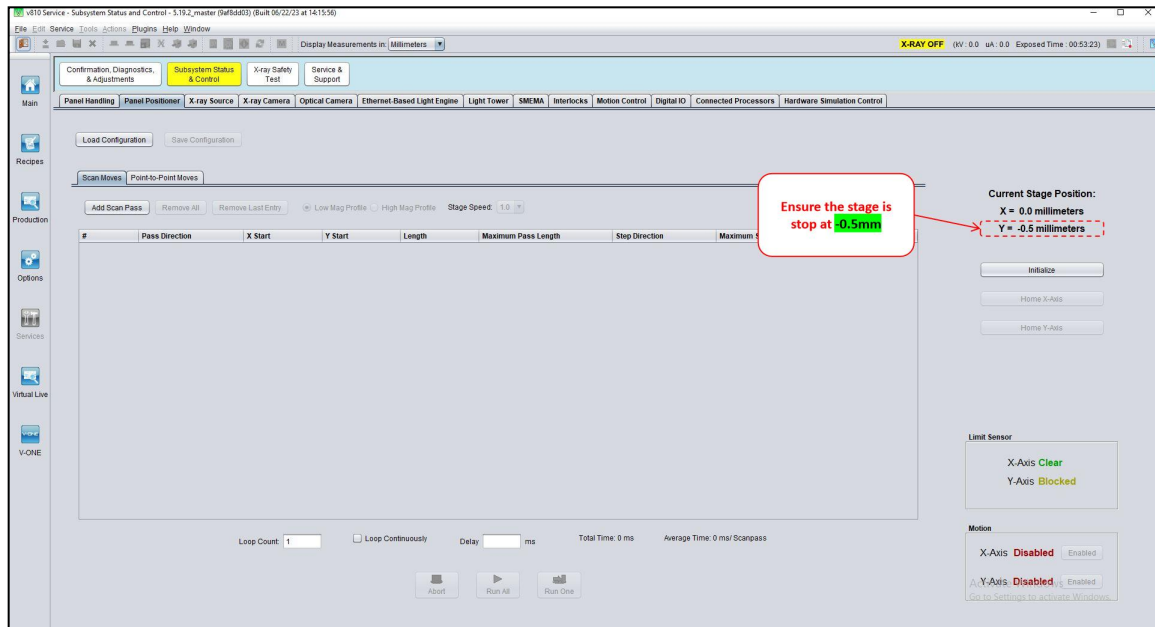


Figure 34: Move Y-Axis Stage to Current Stage Position -0.5mm (Sample Image)

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42. **Y-Axis Cover Plate Assembly:** Carefully assemble back the Y-Axis cover plate

Note: If applicable.

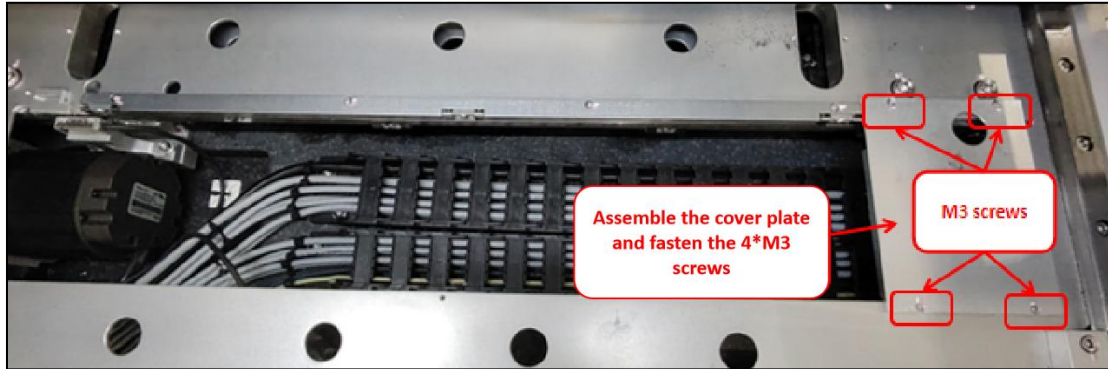


Figure 37: Y-Axis Cover Plate

43. **Validation of Machine Performance After Adjustment and Confirmation:** Machine performance validation is required before release back to customer. Refer to “**Validation of Machine Performance After Alignment for Sensors**” procedure in last page.

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Home and Limit Sensors Alignment for Panasonic Drive

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X-Axis Home and Limit Sensor Alignment for Panasonic Drive

44. **Stop RMS and Close V810:** Navigate to ‘Services > Subsystem Status & Control > Connected Processors’ and click “Stop RMS”. Close the V810 upon RMS have been stopped.

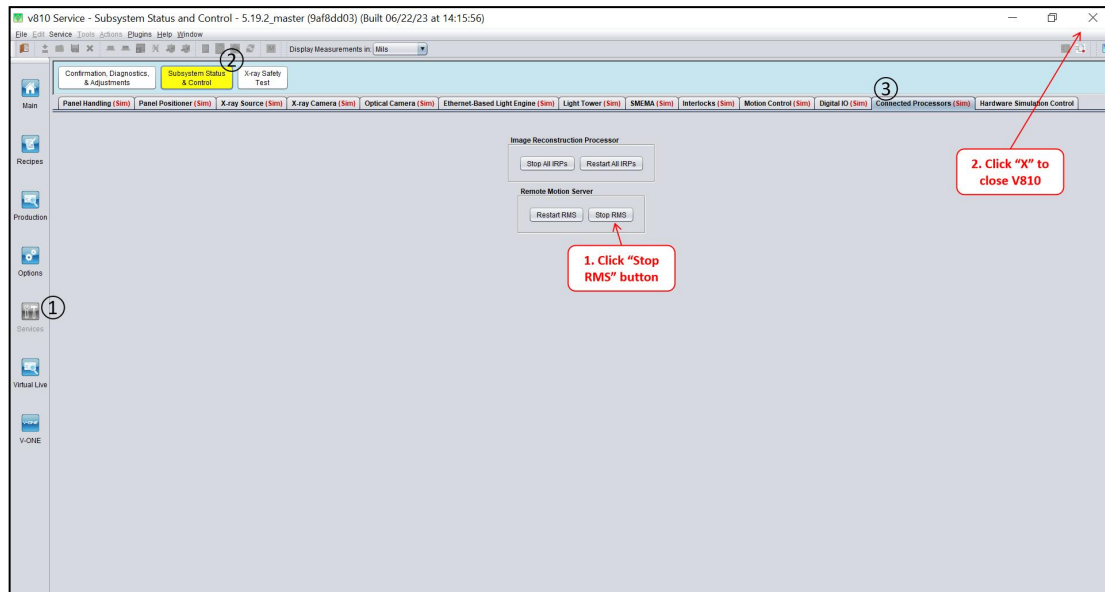


Figure 38: Stop RMS (Sample Image)

45. **Connection Establishment:** Unplug the LAN cable from port X2A and connect the USB type A to mini-B cable from Atom PC to the driver port X1. Then, power down and restart SSCA.

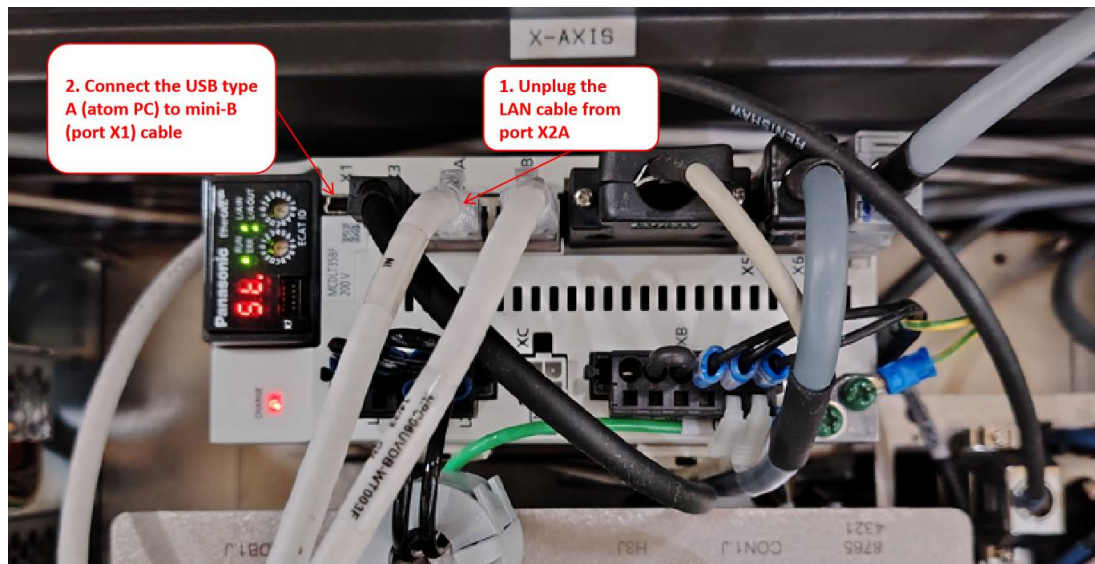


Figure 39: Connection Establishment (Sample Image)

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46. **Open PANATERM Application:** Remote into Atom PC, Open the PANATERM software and connect to the driver. Then select Other > Object Editor > Read from the file > OK.

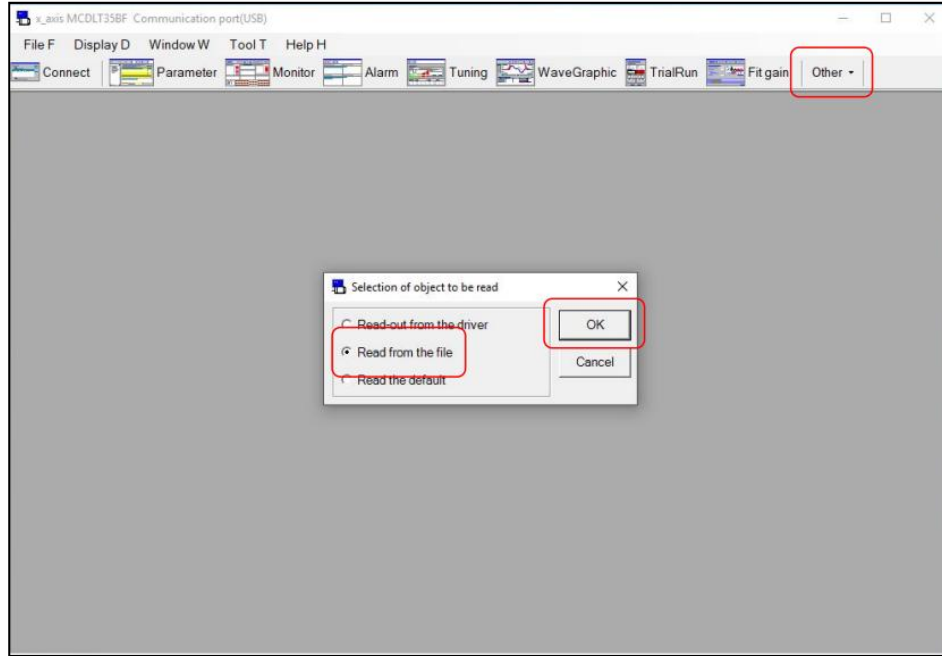


Figure 40: Panaterm Application (Sample Image)

47. **Open Path Window Prompt Out:** Navigate to the path 'C:\Program Files\AXI System\v810\5.0**Config\motionControl\motionDrive\ethercat\Panasonic_A6BF'. Then, choose the correct file for configuration to update.

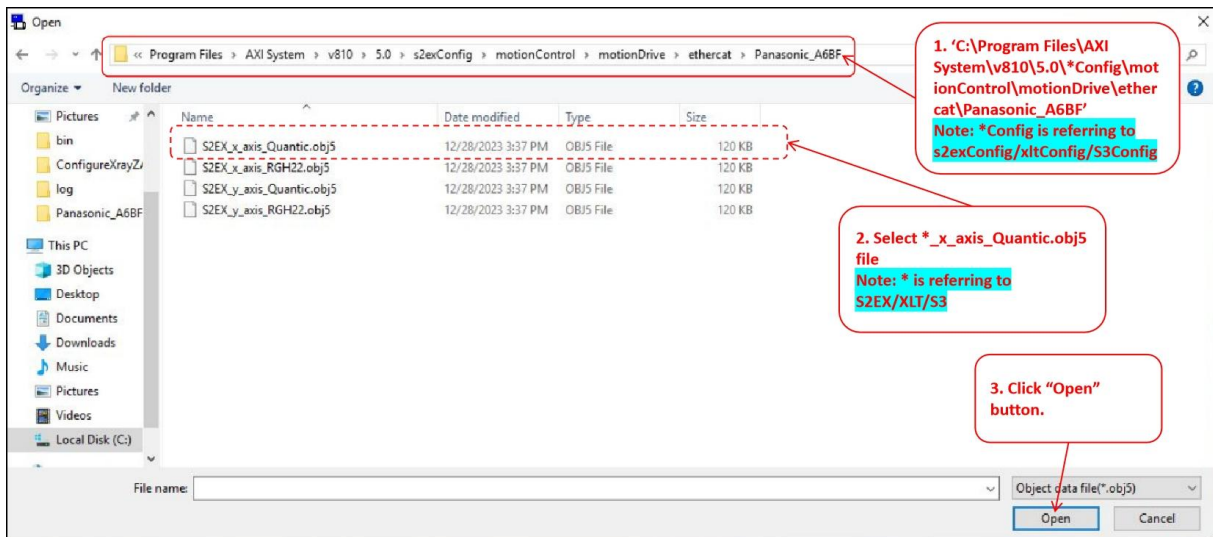


Figure 41: Open Window (Sample Image)

	Work Instruction AXI V810 XY-Axes Quantic Home and Limit Sensors Upgrade Alignment Procedure (AKD, S200, Panasonic)	Doc. Revision 00	
		Effective Date 15 Aug 2024	

48. **Writing the latest configuration to EEPROM:** Click “Trans” button and click “EEP” button to writing the configuration to EEPROM. Then, click “OK” button upon the window for Writing to EEPROM is prompt out.

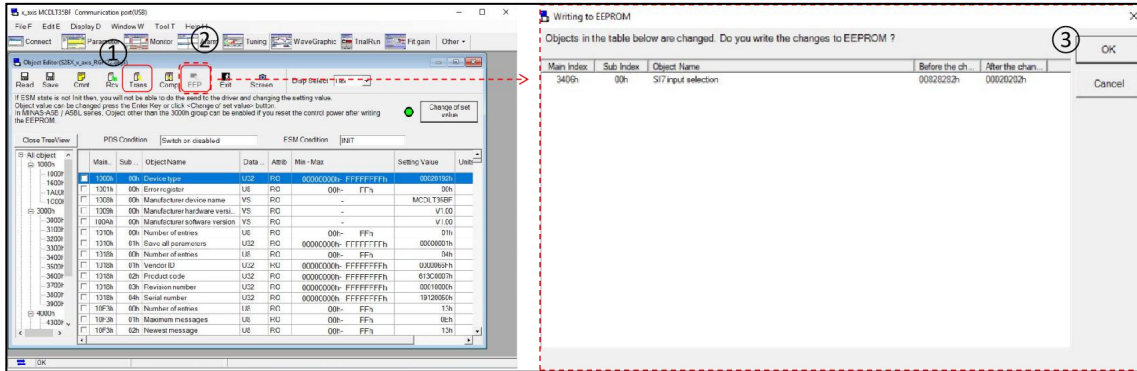


Figure 42: Writing to EEPROM (Sample Image)

49. **Status Verification for Pin12(SI7):** Navigate to “Other” and select “Pin Assign”. Then, validate the status for pin 12(SI7) as per illustrated.

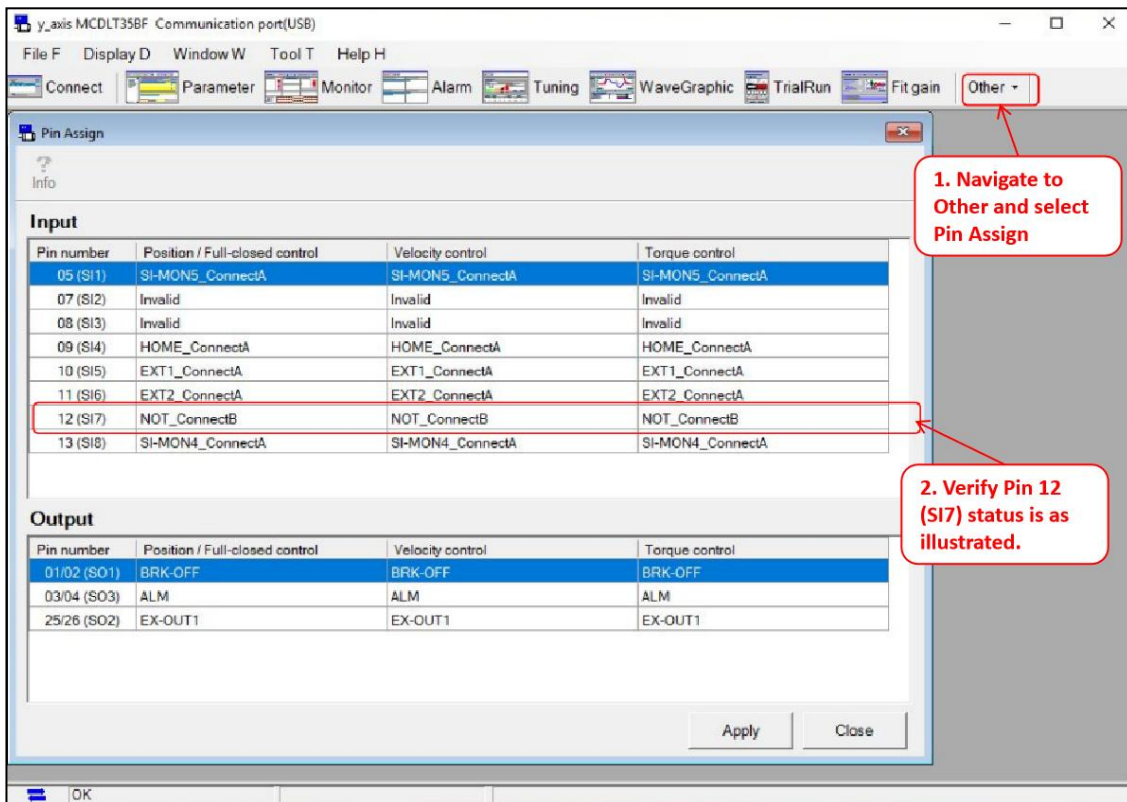


Figure 43: Verification for Pin 12 Status (Sample Image)

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50. **Power off SSCA, Revert the Driver Connection to Original Setup, Power on SSCA:** Power down SSCA and re-connect the LAN cable, unplug the USB type-A to mini-B cable from the Atom PC and Panasonic driver. Power on SSCA once complete revert to original setup.

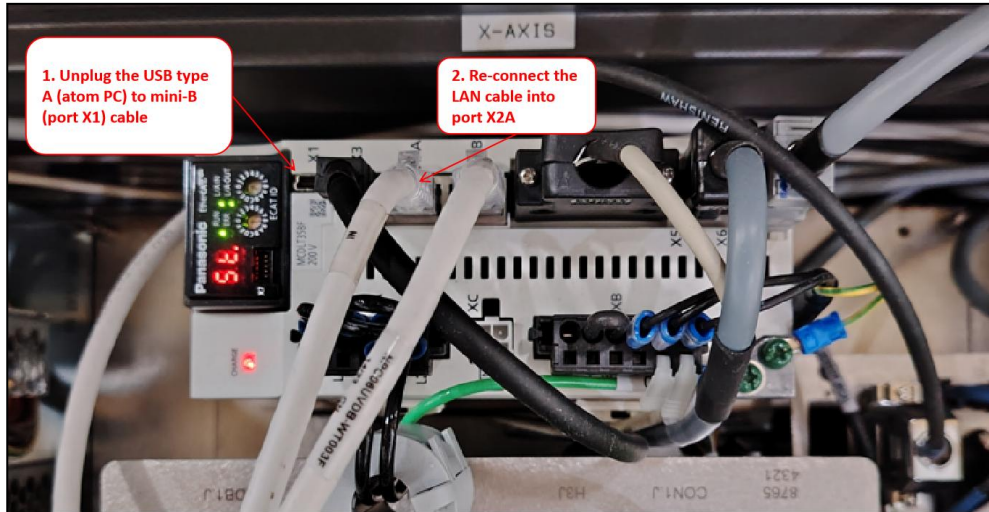


Figure 44: Revert to Original Connection (Sample Image)

51. **Unplug the LAN Cable Labeled with Slice IO from the Atom PC:** Disconnect the LAN cable from Atom PC that is connecting to the slice IO.

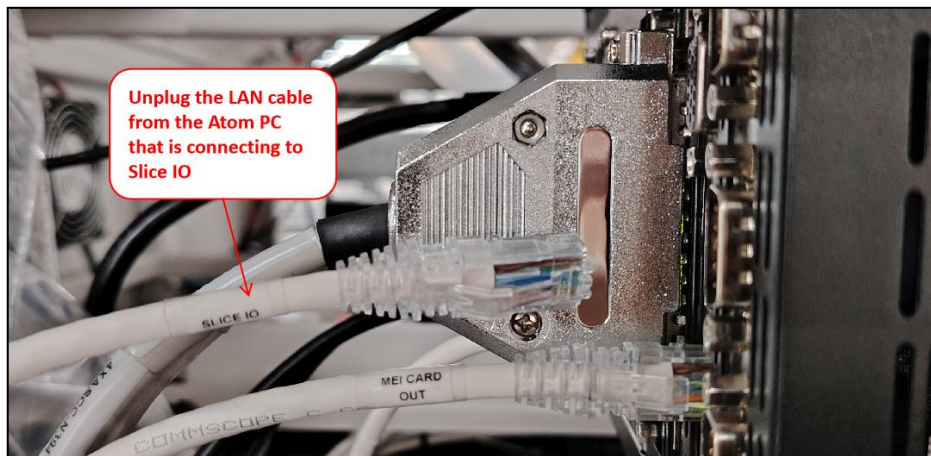


Figure 45: Unplug LAN cable from Atom PC

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52. **Launch Remote Desktop Connection:** Go to search, key in remote desktop connection and launch the application.

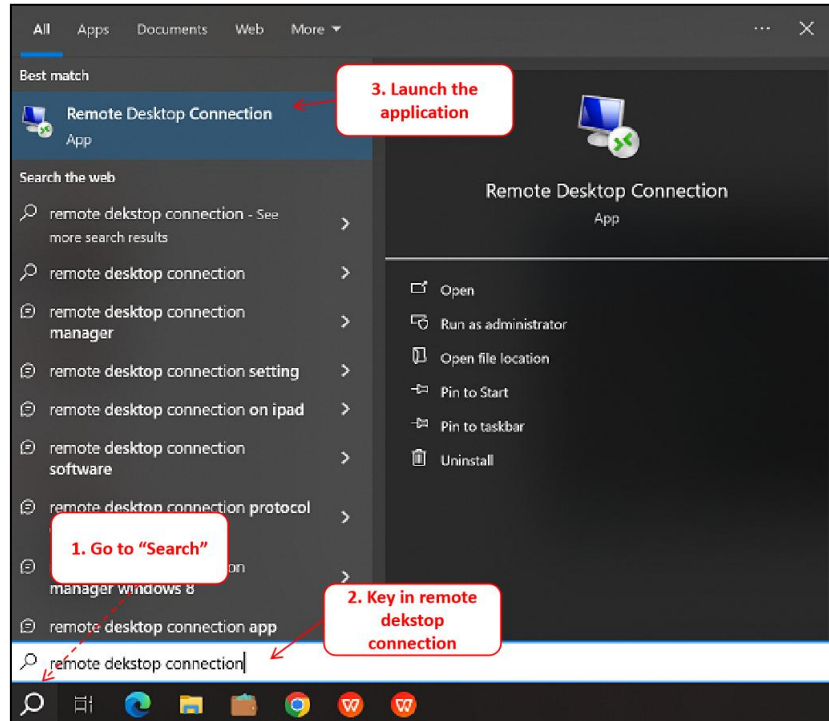


Figure 46: Launch the Remote Desktop Connection (Sample Image)

53. **Connect to Atom PC (192.168.128.70):** Key in 192.168.128.70 at the computer address input box, hit enter to “Connect”.



Figure 47: Remote Desktop Connection (Sample Image)

	Work Instruction	Doc. Revision	00
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54. **Launch the PanasonicHomeSensorAdjustmentTools.exe Program:** Navigate to Desktop and look for PanasonicHomeSensorAdjustmentTools.exe application. Double click the icon to open the program.

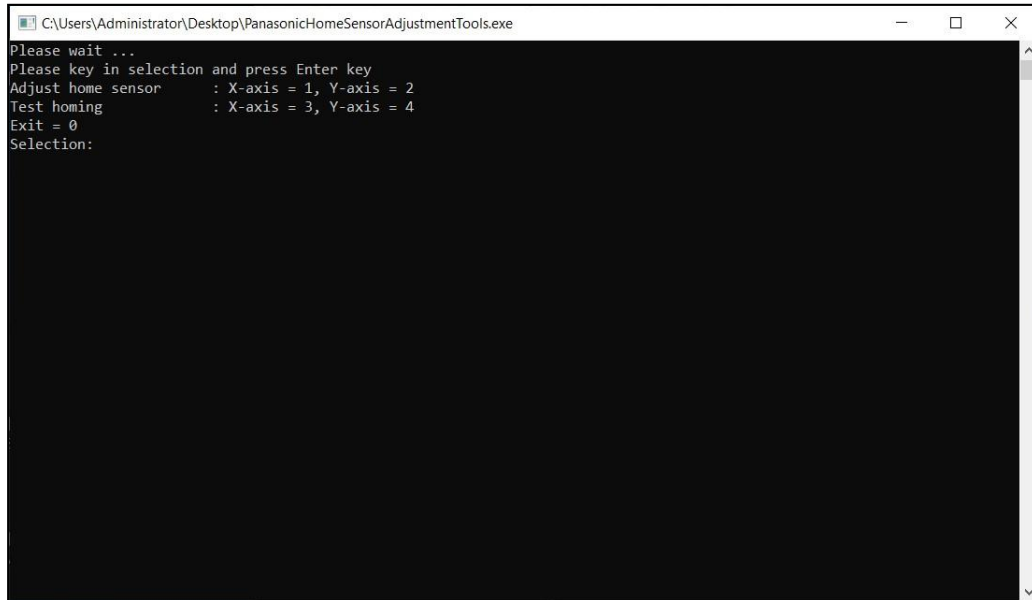


Figure 48: Launch PanasonicHomeSensorAdjustmentTools

55. **Launch the “Common Motion Utility” Application:** Go to Windows Search. Input “Common Motion Utility” and launch the application. Wait patiently for the application to startup.

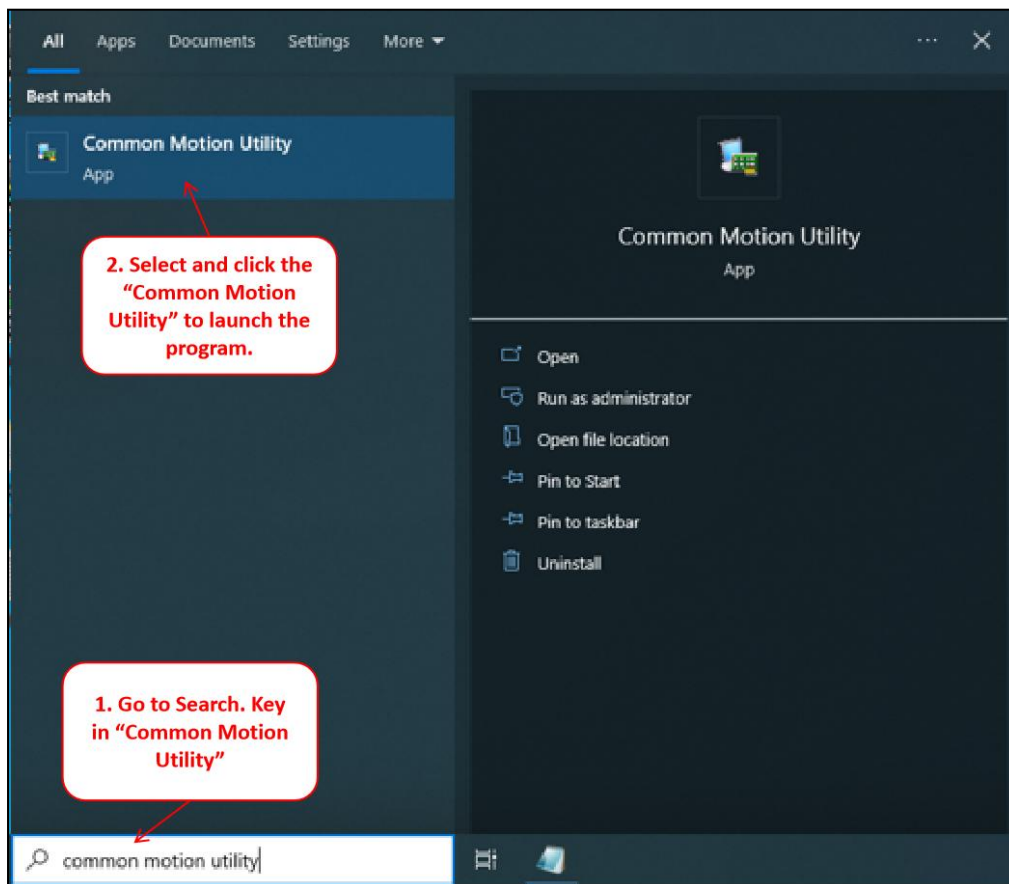


Figure 49: Launch Common Motion Utility Application

	Work Instruction	Doc. Revision	00
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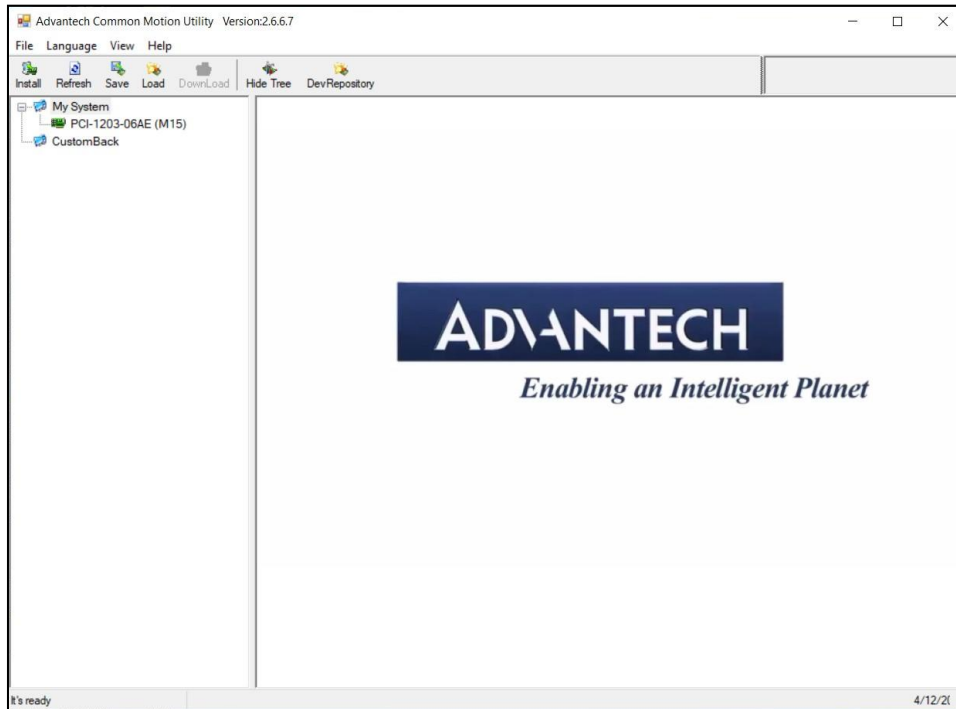


Figure 50: GUI for Common Motion Utility Startup

56. **Expand the list for PCI-1203-06AE (M15):** Double click the PCI-1203-06AE (M15) to expand its selection list. Under the Motion Ring node, there are 2 items as per below.

56.1 0x001: MCDLT35BF: For X-axis home position and limit sensor adjustment

56.2 0x002: MCDLT35BF: For Y-axis home position and limit sensor adjustment

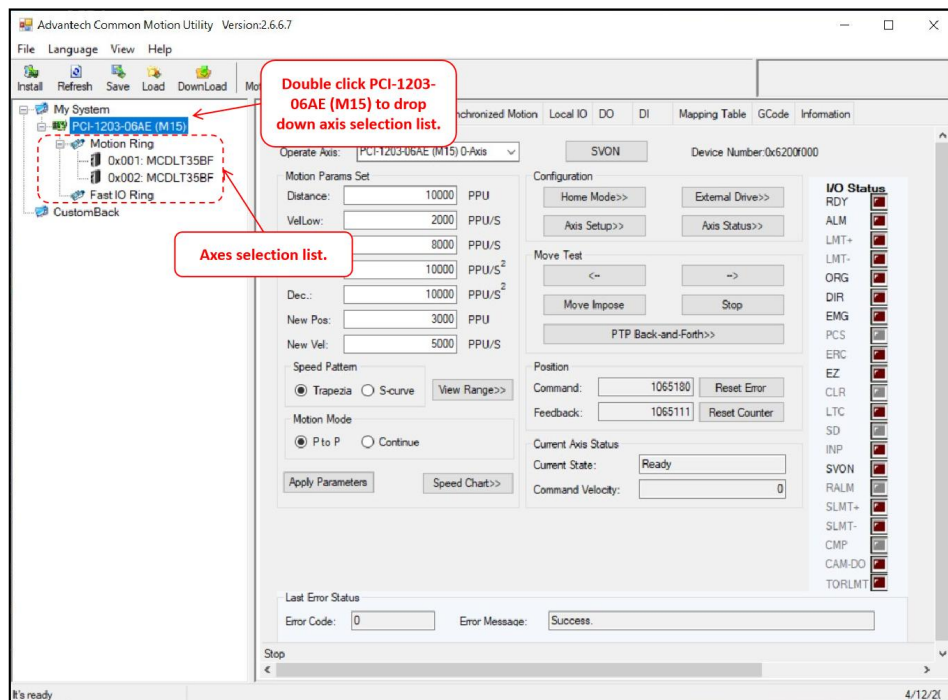


Figure 51: List Expansion for PCI-1203-06AE (M15)

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57. **Axis Selection:** Select the 0x001: MCDLT35BF to proceed home position adjustment for x-axis.

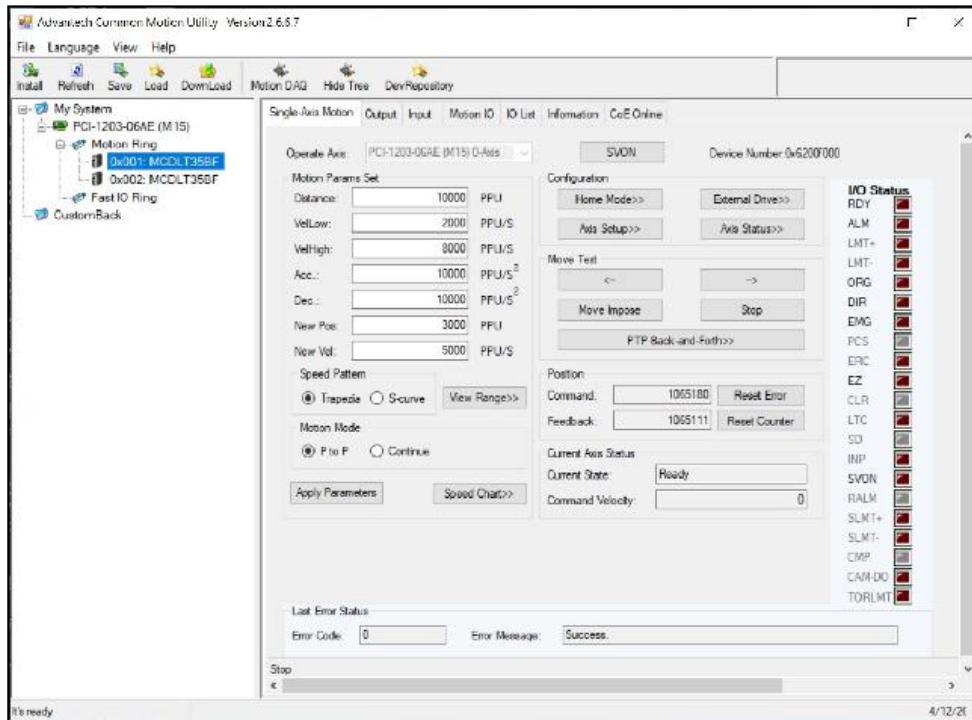


Figure 52: Axis Selection for Home Position Adjustment

58. **Ensure LMT- is Triggered at Hard Stop Position:** Upon axis selection, move the specific axis stage to hard stop position and ensure the LMT- is in triggered mode. Else, adjust the limit sensor for signal to be triggered.

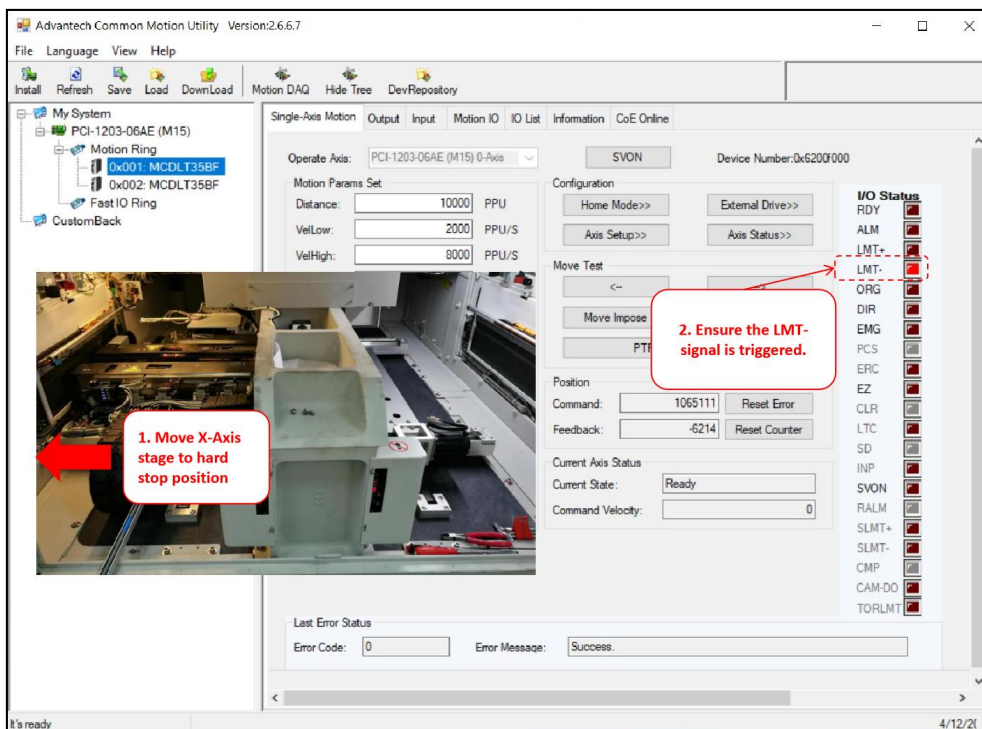


Figure 53: LMT- Signal Triggering Status

	Work Instruction	Doc. Revision	00
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59. **X-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the X-Axis stage away from hard stop position.

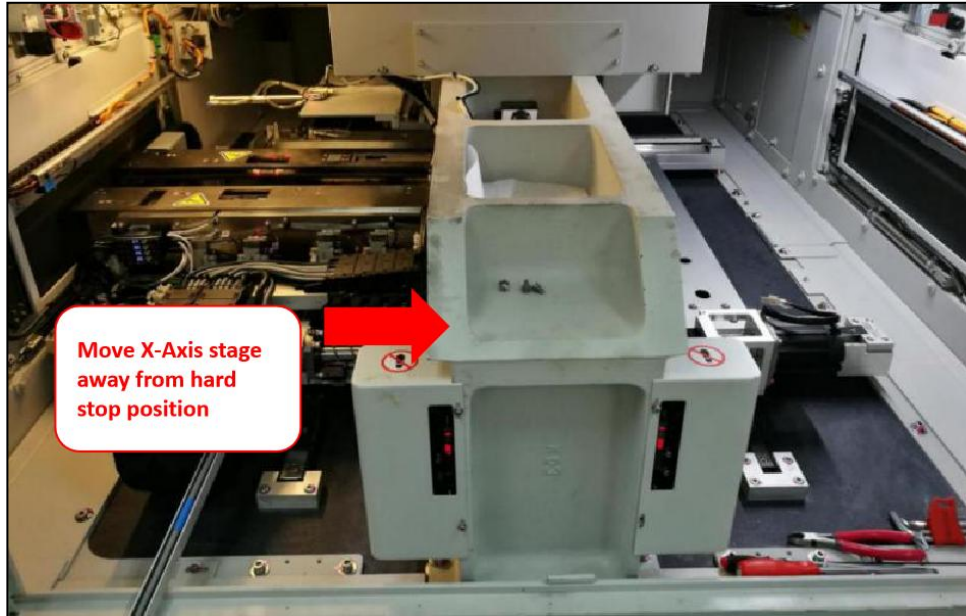


Figure 54: Move X-Axis Stage Away From Hard Stop

60. **Perform Homing:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 1 for x-axis to go home position.

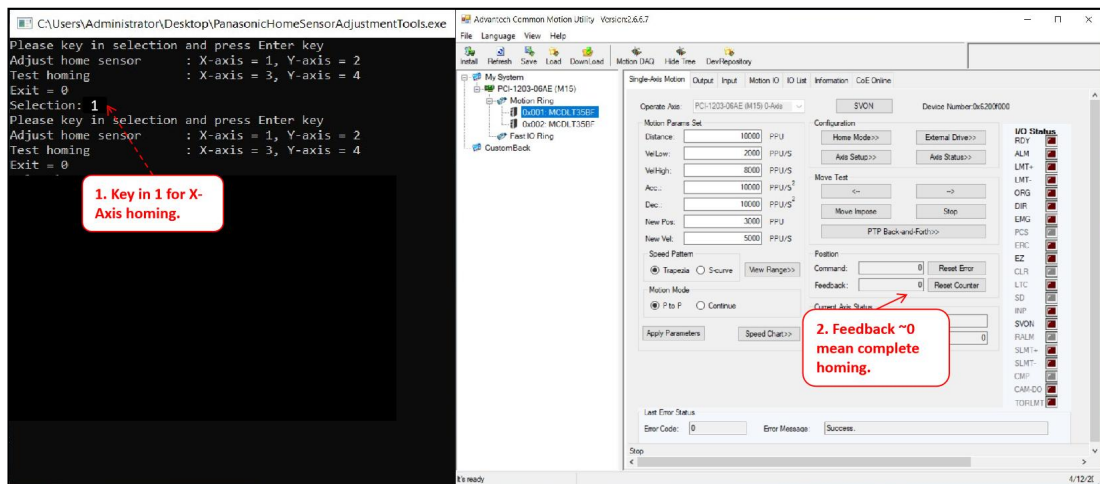


Figure 55: Perform Homing

	Work Instruction	Doc. Revision	00
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61. **Move the Stage to Hard Stop Position:** Move the X-stage to the hard stop position.

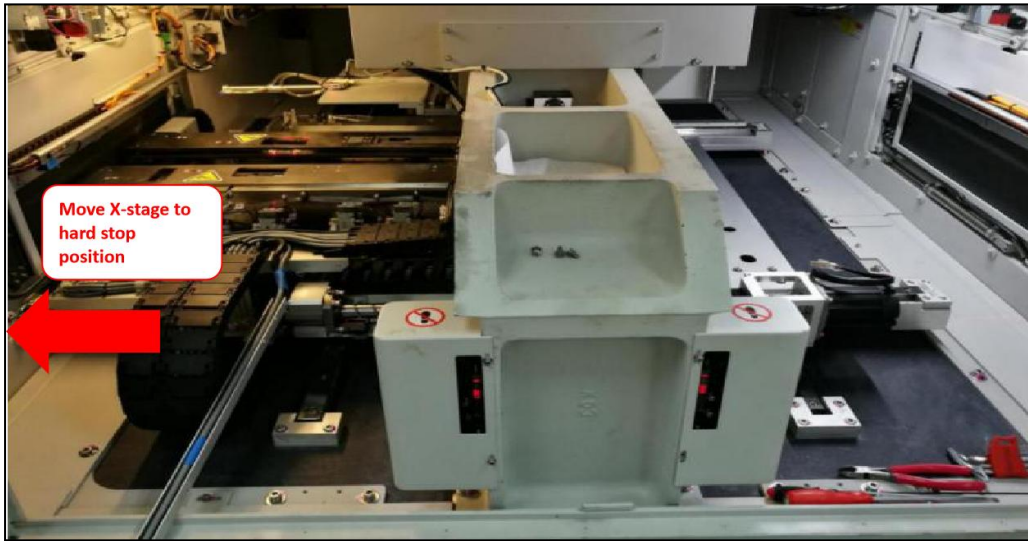


Figure 56: Move Stage to Hard Stop Position

62. **Validate the Feedback Position is 4980-5180:** At the hard stop position, validate the feedback position. If it is not within acceptable range of 4980-5180, please proceed with Step 63. Else, proceed to Step 67.

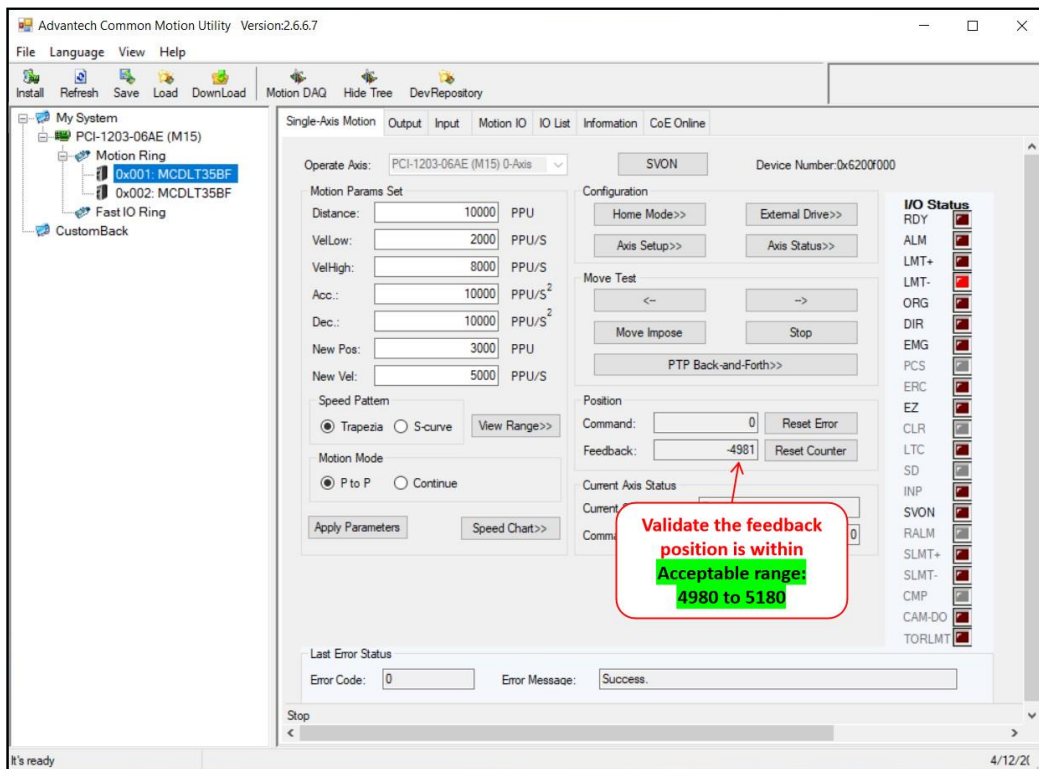


Figure 57: Validation of Feedback Position

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63. **Loosen Spring Clamp:** Loosen all the screws for the spring clamp.

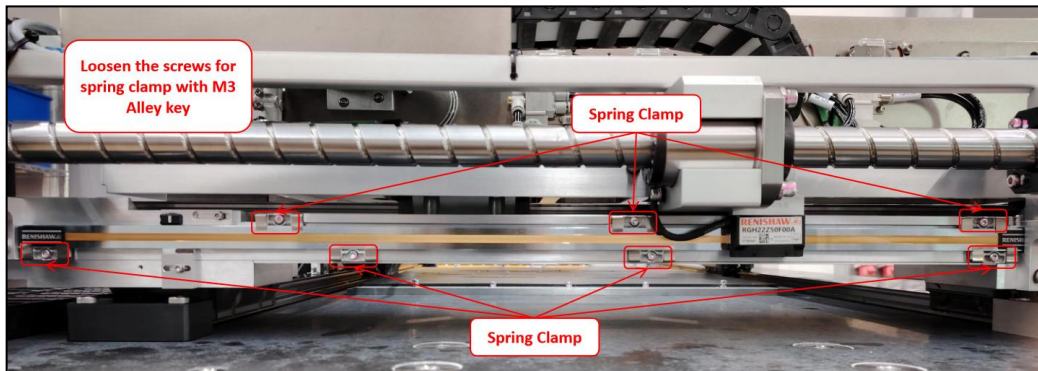


Figure 58: Quantic Scale Spring Clamps (Sample Image)

64. **Quantic Scale Positioning Adjustment:** Gently move the Quantic scale. While moving the Quantic scale, check the “Feedback Position” in Common Motion Utility and ensure the reading fall in the range 4980 to 5180.

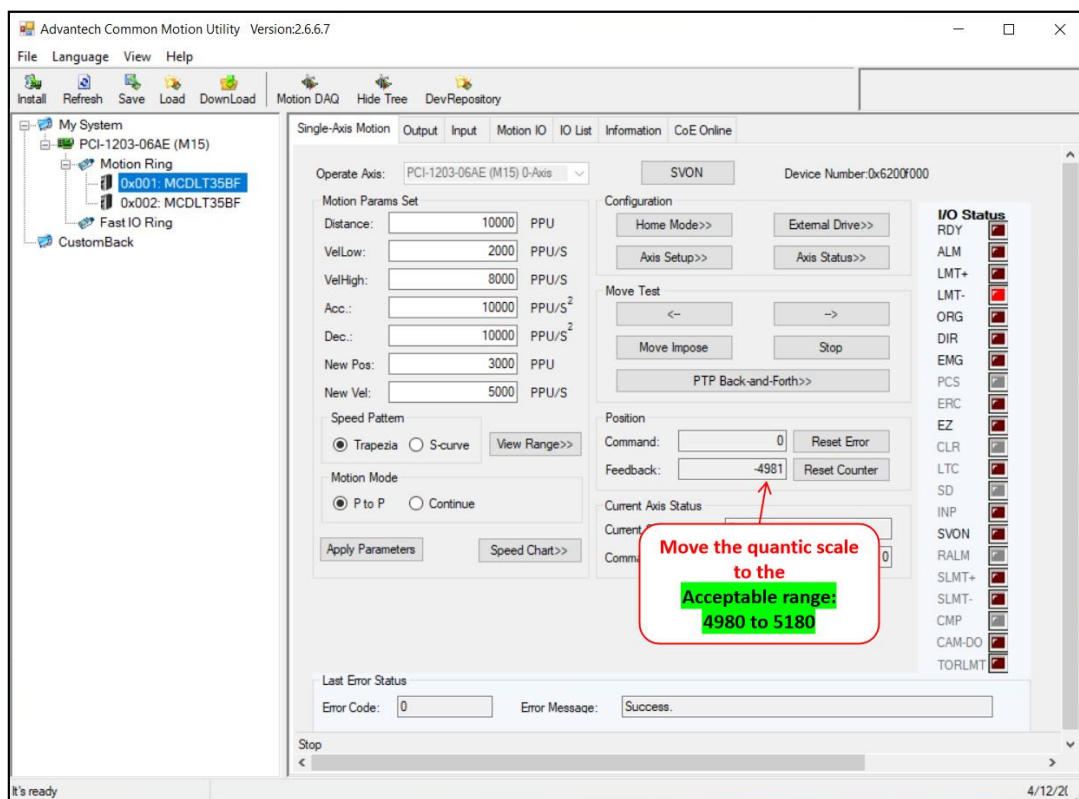


Figure 59: Quantic Scale Adjustment by Monitoring the Feedback Position

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65. **Tighten the Spring Clamps:** Upon completion of home position alignment, tighten all screw for the spring clamps.

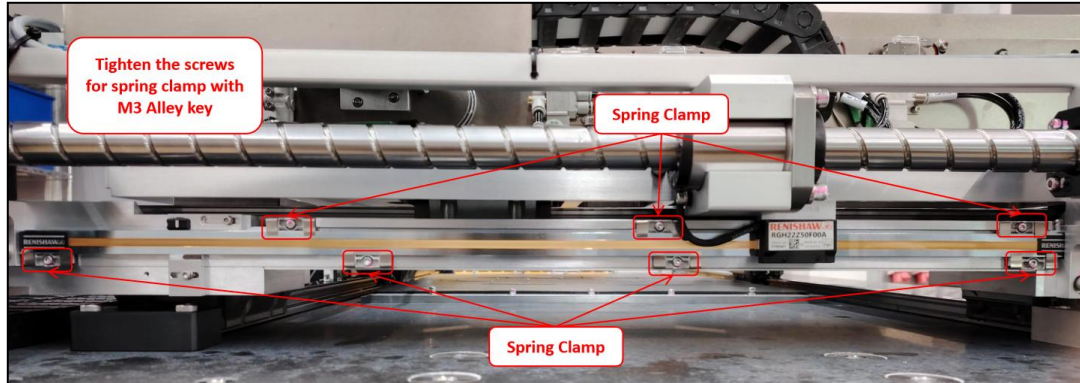


Figure 60: X-Axis Quantic Scale Spring clamp (Sample Image)

66. **Home Sensor Positioning Offset:** Move the position of the home sensor to cross the engraved line mark at the top side of the Quantic scale.

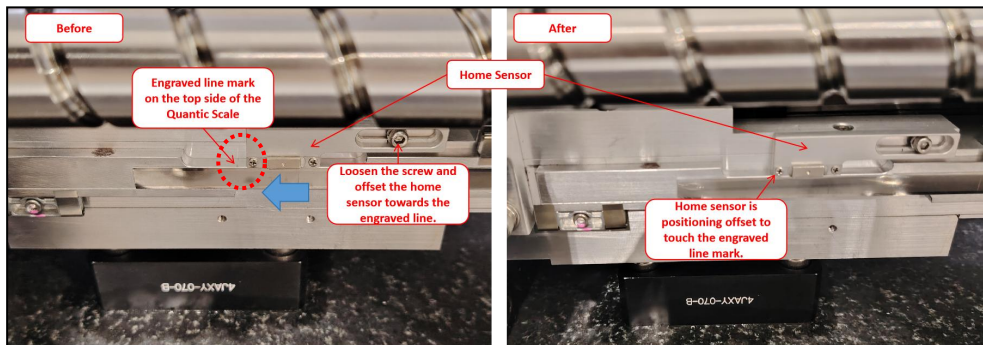


Figure 61: X-Axis Home Sensor Offset

67. **X-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the X-Axis to the middle of the stage.

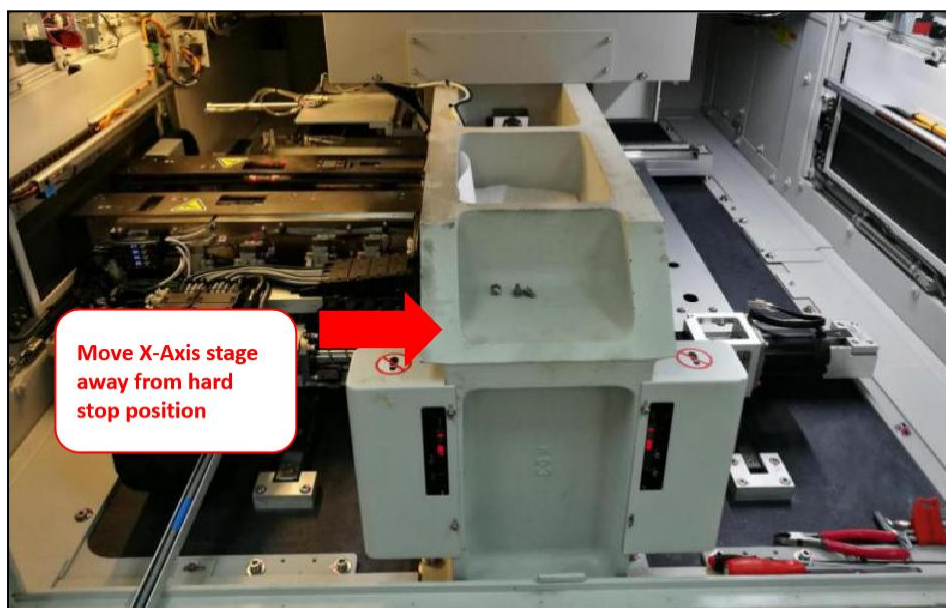


Figure 62: Move X-Axis Stage Away From Hard Stop

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68. **Homing again:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 1 for x-axis to go home position.

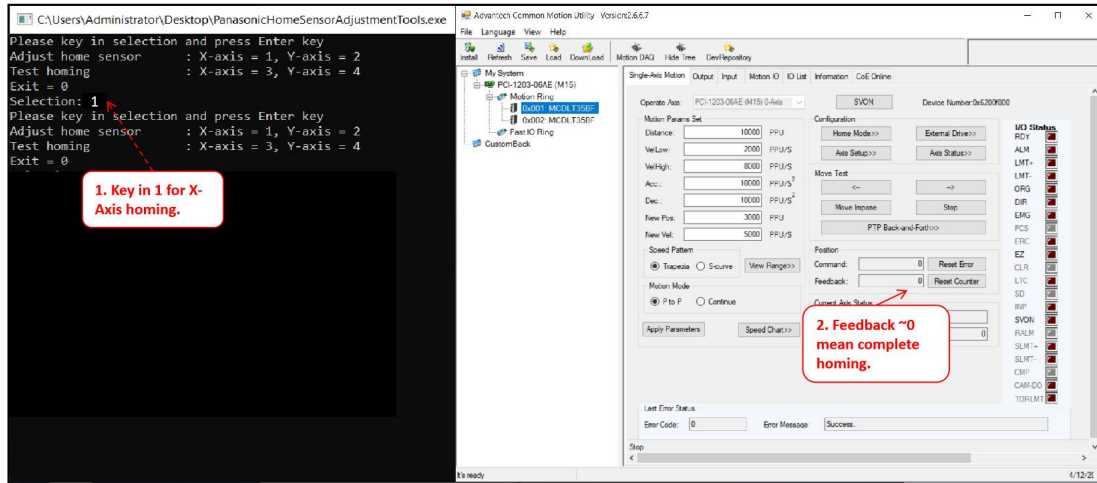


Figure 63: Perform Homing

69. **Move the Stage towards Hard Stop direction until it hit 1755 ± 1 :** Move the stage towards the hard stop direction and ensure the feedback position is within the range of 1754 to 1756.

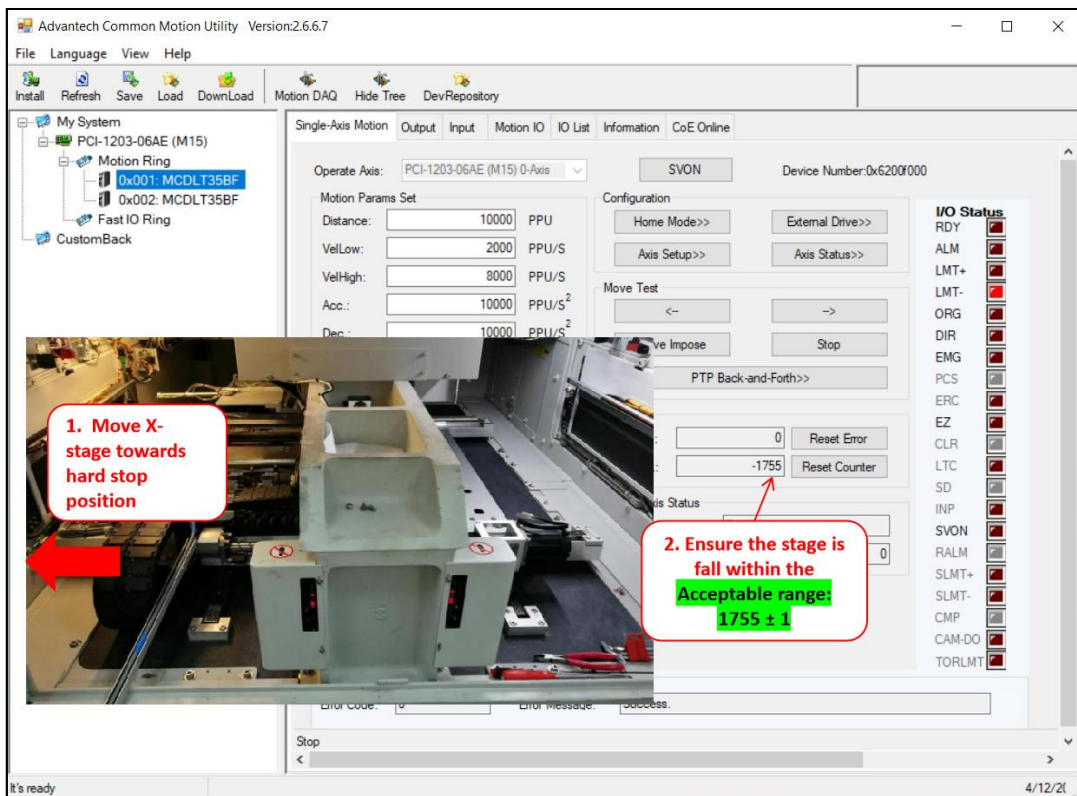


Figure 64: Move X-Axis Stage to Feedback Position 1755 (Sample Image)

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70. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

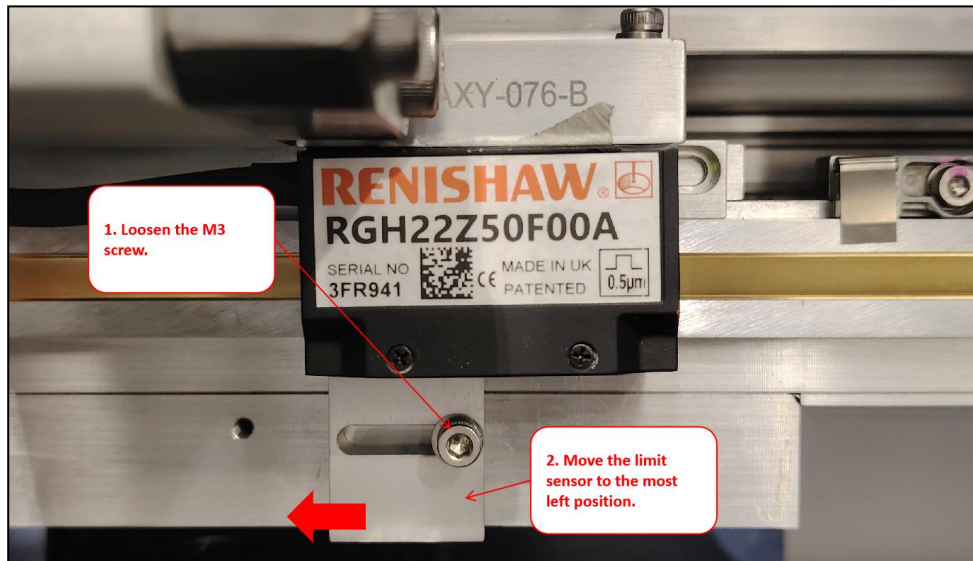


Figure 65: Move Limit Sensor to the Most Left Position

71. **Adjust the limit sensor to right hand side until LMT- is triggered:** Slowly move the limit sensor towards the right hand side position until the LMT- signal is triggered.

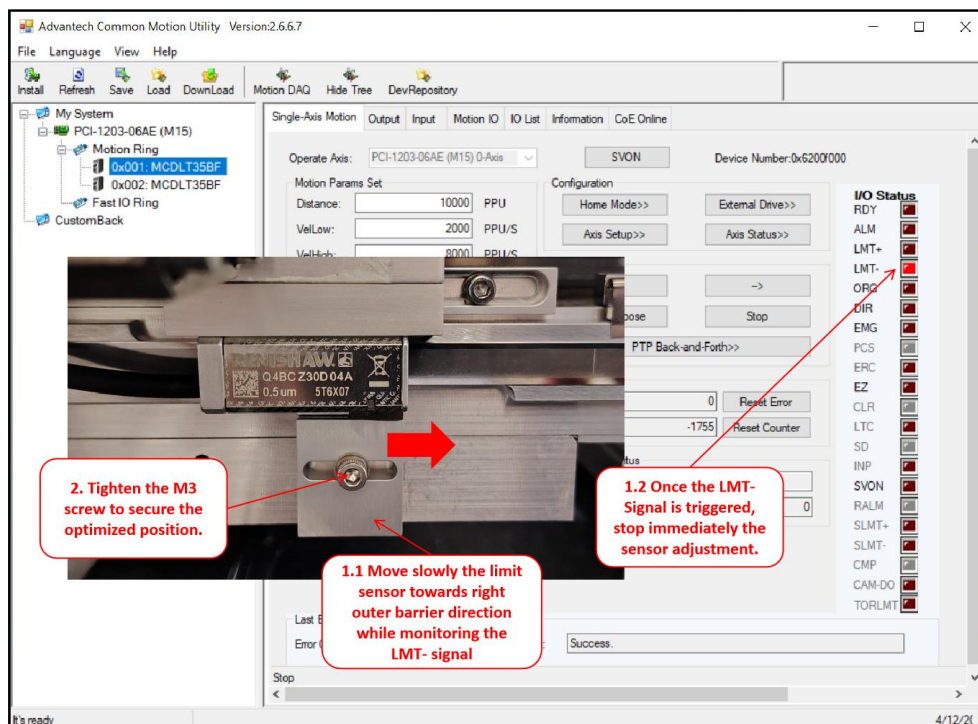


Figure 66: Positioning Limit Sensor to Triggered State

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72. **Proceed to Test Homing for Final Verification:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 3 for x-axis to test homing result.

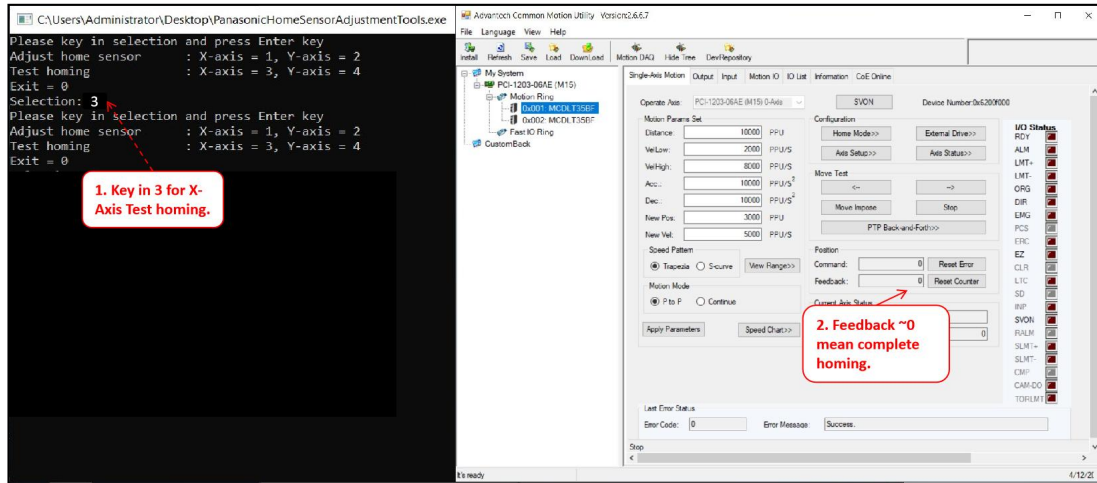


Figure 67: Perform Homing for Final Verification

73. **Re-connect the LAN cable labeled with Slice IO to the Atom PC:** Re-connect the LAN cable labeled with Slice IO to the Atom PC.



Figure 68: Reconnect the LAN Cable for Slice IO

74. **Go to Main Server and open v810:** Close remote desktop connection and launch the v810 program.
75. **Panel Positioner Initialization:** Navigate to 'Service > Subsystem Status & Control > Panel Positioner'. Click "Initialize".

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76. **Verify the “Current Stage Position” for X-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the X stage to the hard stop position and verify the home sensor reading is within 3.08mm to 3.28mm. If not within the range, route back to Step 44. Else, proceed to the next step.

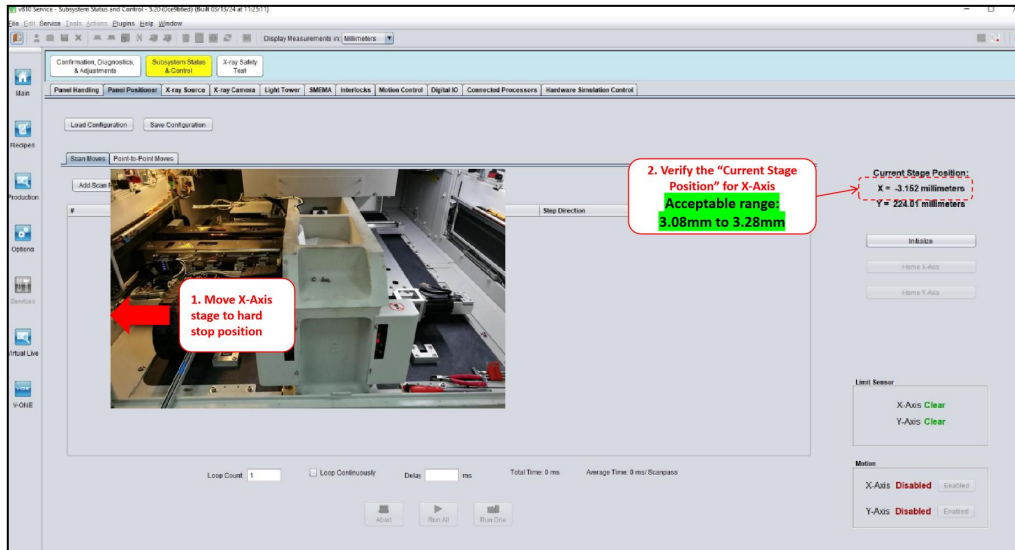


Figure 69: Verification of Current Stage Position for X-Axis

77. **Proceed to Validate Machine Performance:** Validation of machine is required before release the machine to production. Refer to “**Validation of Machine Performance After Alignment for Sensors**” procedure in last page.

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78. **Stop RMS and Close V810:** Navigate to ‘Services > Subsystem Status & Control > Connected Processors’ and click “Stop RMS”. Close the V810 upon RMS have been stopped.

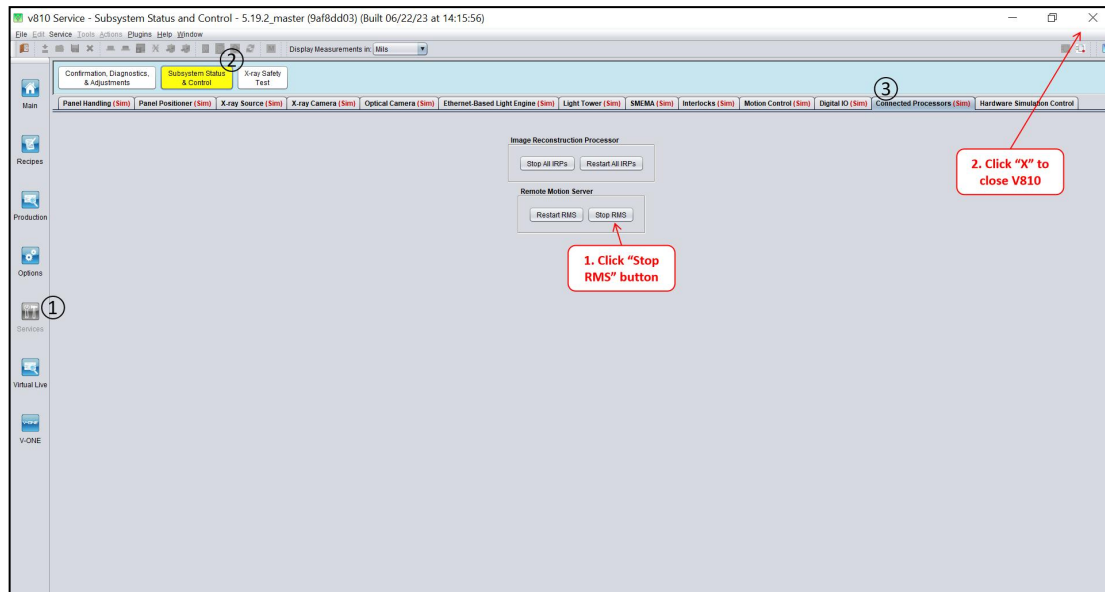


Figure 70: Figure :Stop RMS (Sample Image)

79. **Connection Establishment:** Unplug the LAN cable from port X2A and connect the USB type A to mini-B cable from Atom PC to the driver. Then, power down and restart SSCA.

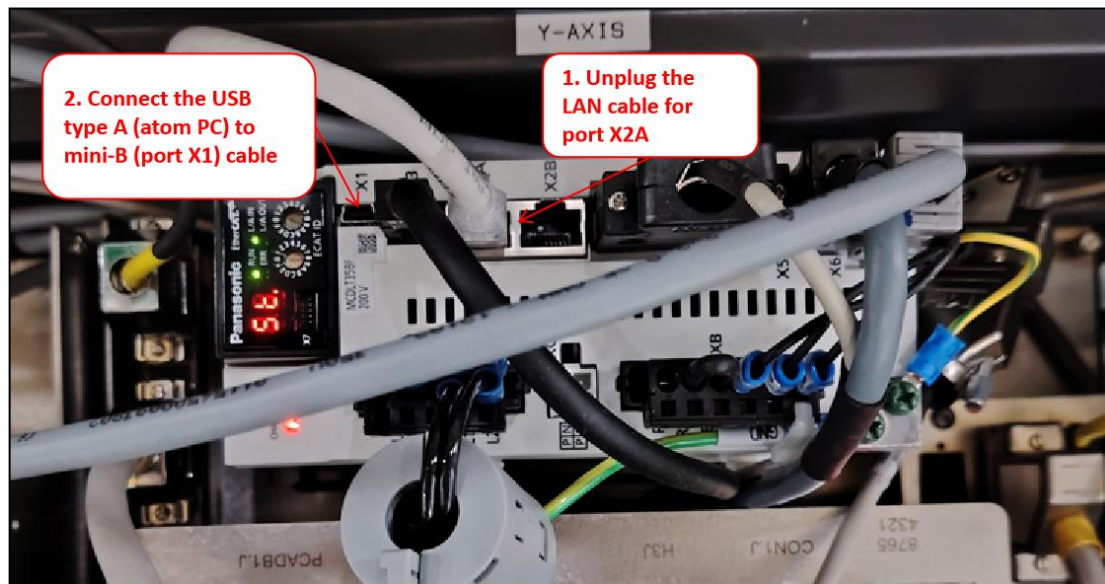


Figure 71: Figure :Connection Establishment (Sample Image)

80. **Open PANATERM Application:** Open the PANATERM software and connect to the driver. Then select Other > Object Editor > Read from the file > OK.

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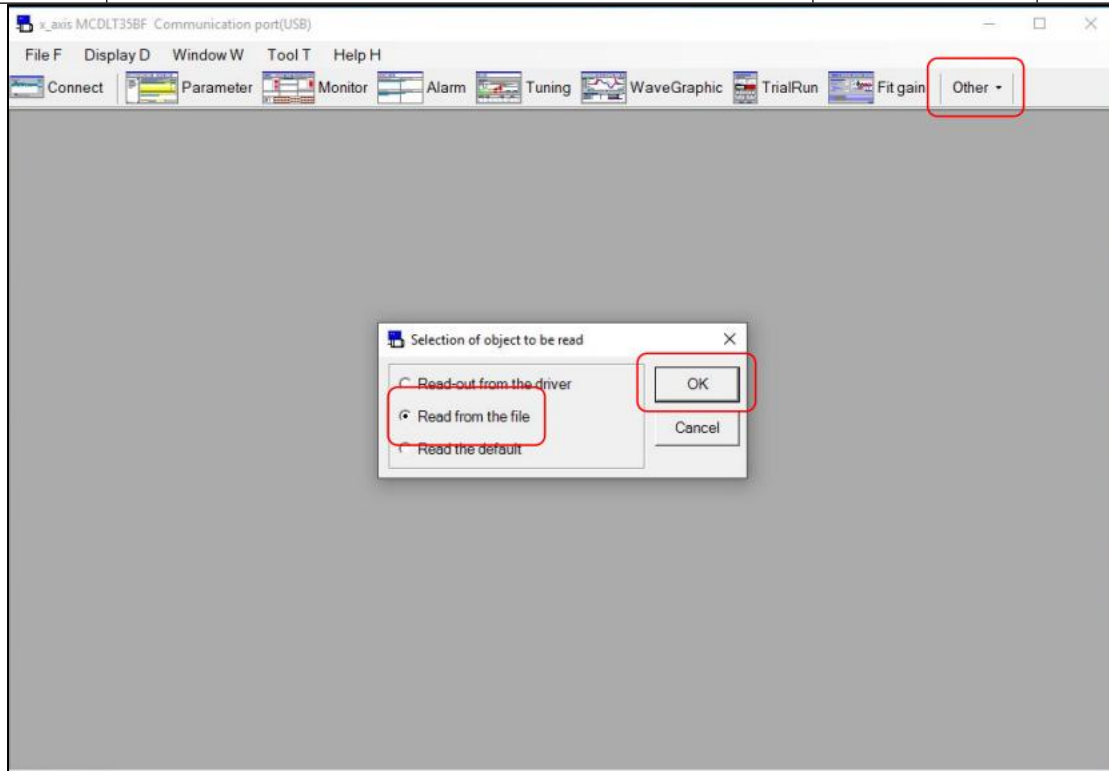


Figure 72: Figure :Panaterm Application (Sample Image)

81. **Open Path Window Prompt Out:** Navigate to the path ‘C:\Program Files\AXI System\v810\5.0*Config\motionControl\motionDrive\ethercat\Panasonic_A6BF’. Then, choose the correct file for configuration to update.

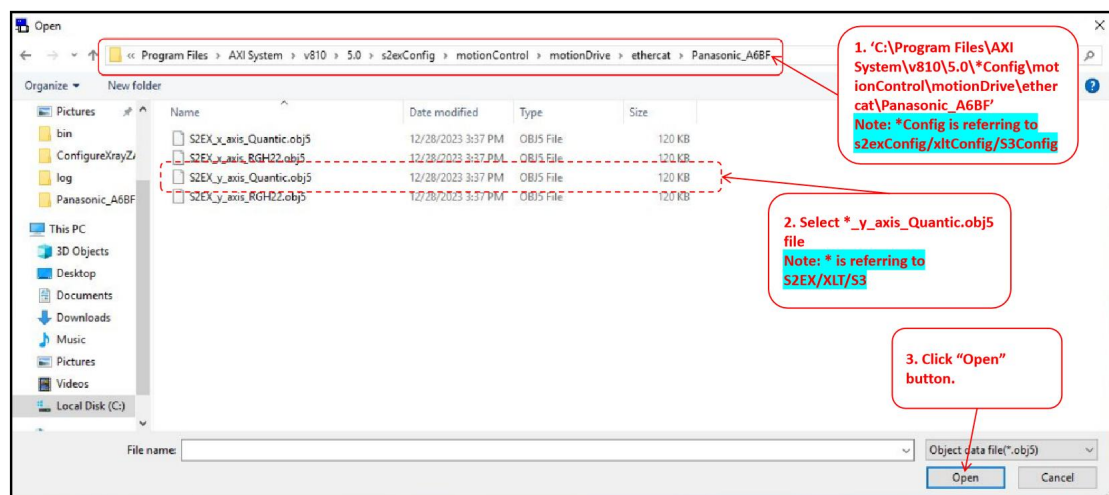


Figure 73: Figure :Open Window (Sample Image)

82. **Writing the latest configuration to EEPROM:** Click “Trans” button and click “EEP” button to writing the configuration to EEPROM. Then, click “OK” button upon the window for Writing to EEPROM is prompt out.

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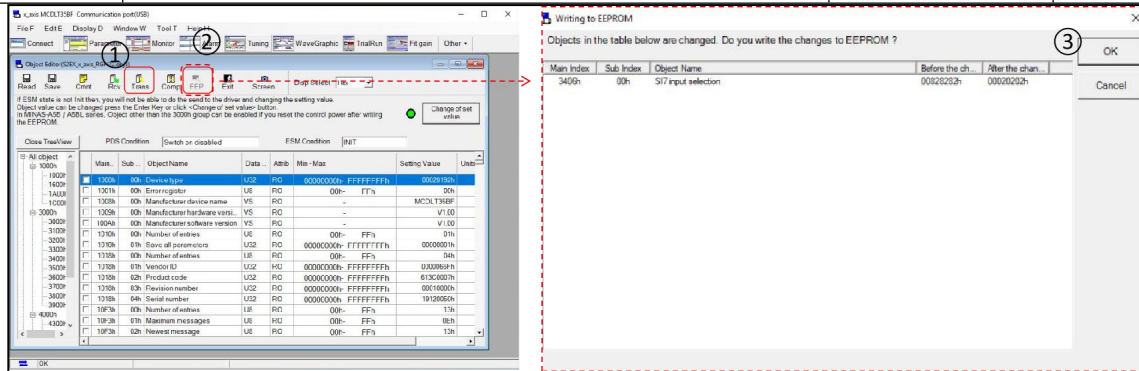


Figure 74: Figure :Writing to EEPROM (Sample Image)

83. **Status Verification for Pin12(SI7):** Navigate to “Other” and select “Pin Assign”. Then, validate the status for pin 12(SI7) as per illustrated.

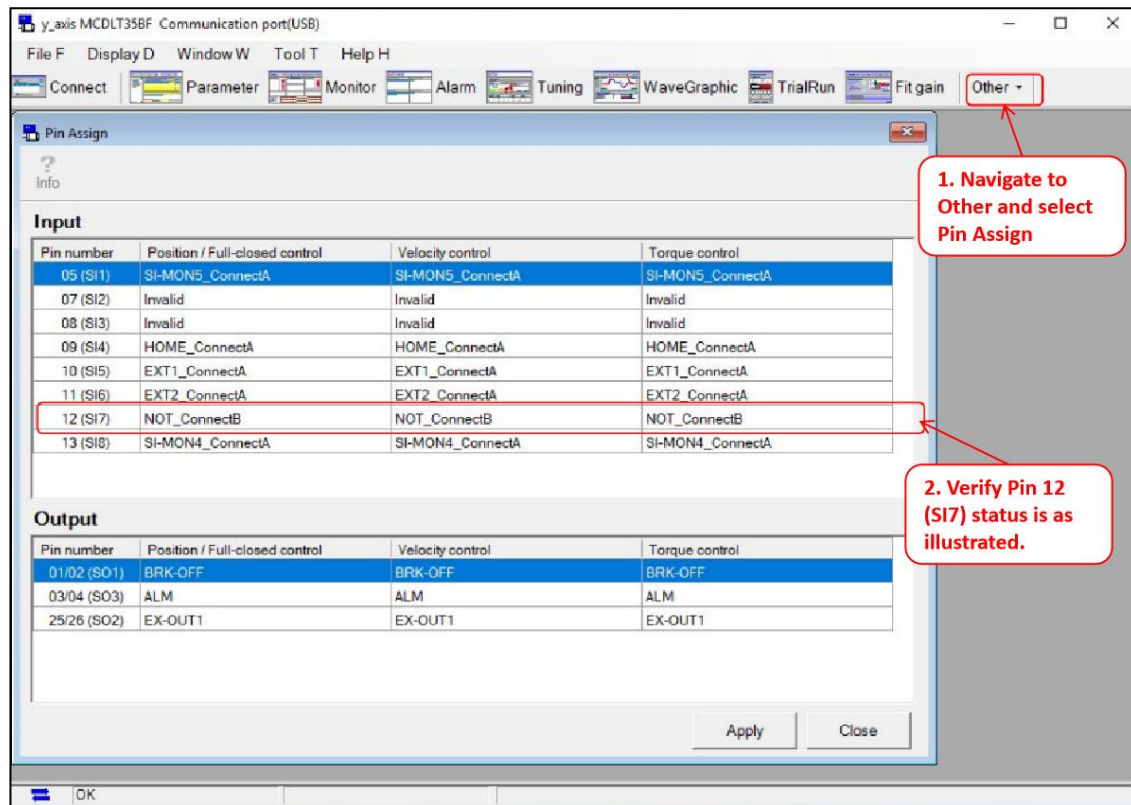


Figure 75: Figure :Verification for Pin 12 Status (Sample Image)

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84. **Power off SSCA, Revert the Driver Connection to Original Setup, Power on SSCA:** Power down SSCA and re-connect the LAN cable, unplug the USB type-A to mini-B cable from the Atom PC and Panasonic driver. Power on SSCA once complete revert to original setup.

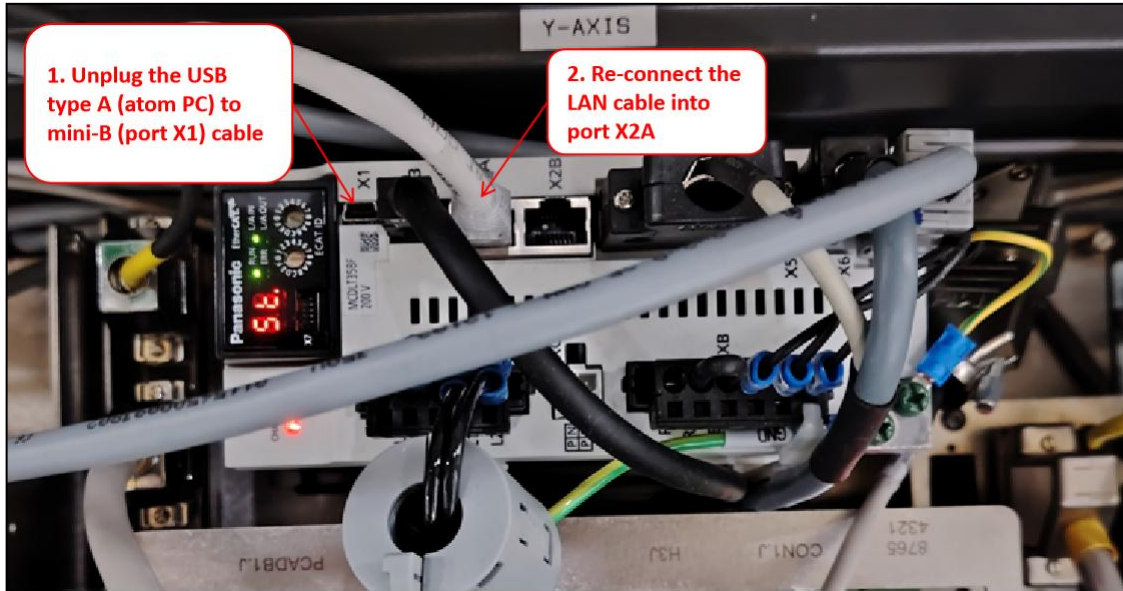


Figure 76: Figure :Revert to Original Connection (Sample Image)

85. **Unplug the Slice IO LAN Cable from Atom PC:** Disconnect the LAN cable that connected to slice IO from Atom PC.

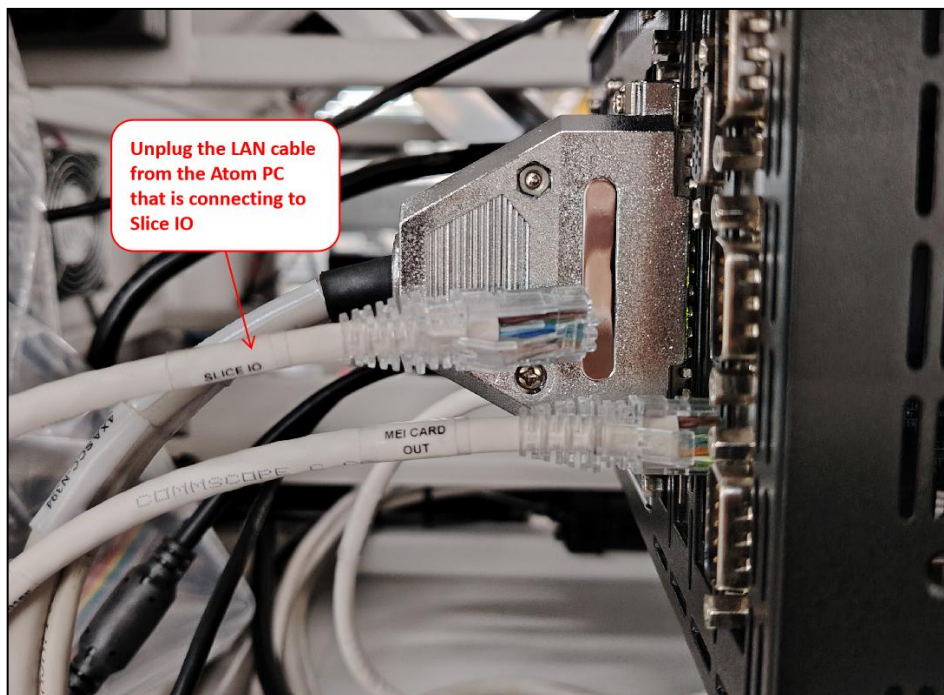


Figure 77: Unplug LAN Cable from the Atom PC

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86. **Launch Remote Desktop Connection:** Go to search, key in remote desktop connection and launch the application.

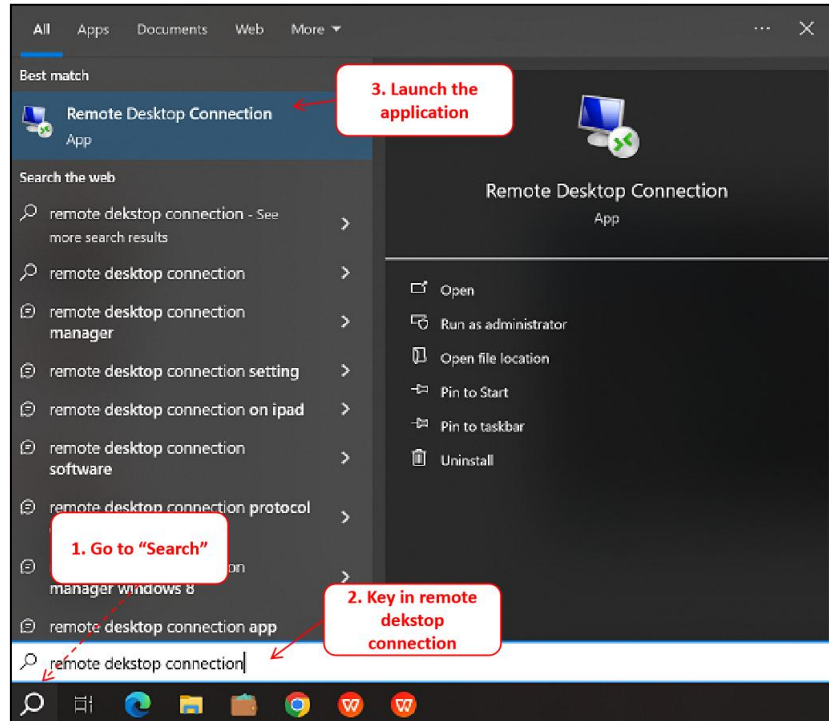


Figure 78: Launch the Remote Desktop Connection (Sample Image)

87. **Connect to Atom PC (192.168.128.70):** Key in 192.168.128.70 at the computer address input box, hit enter to “Connect”.



Figure 79: Remote Desktop Connection (Sample Image)

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88. **Launch the PanasonicHomeSensorAdjustmentTools.exe Program:** Navigate to Desktop and look for PanasonicHomeSensorAdjustmentTools.exe application. Double click the icon to open the program.

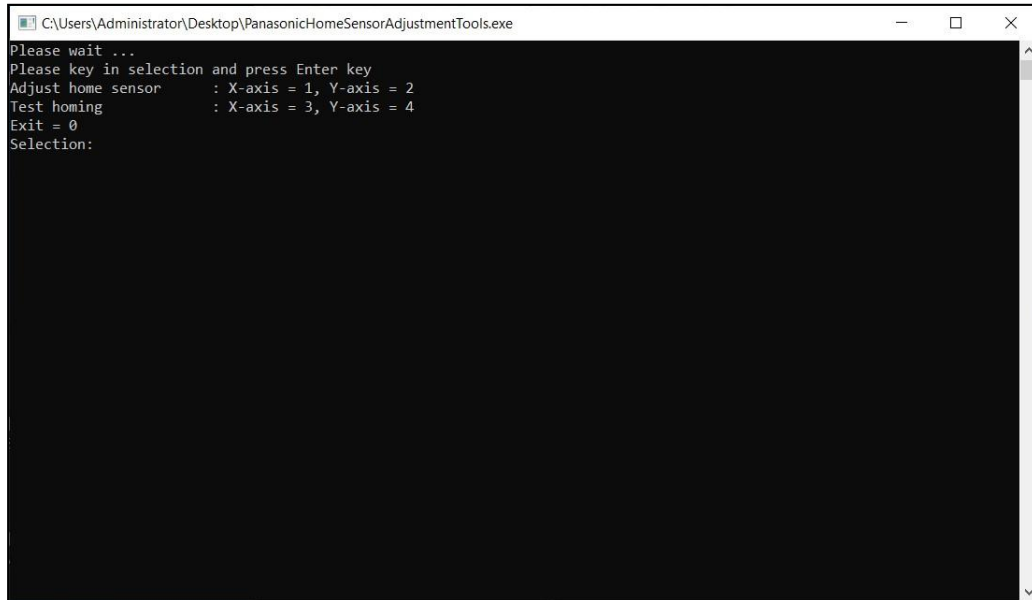


Figure 80: Launch PanasonicHomeSensorAdjustmentTools

89. **Launch the “Common Motion Utility” Application:** Go to Windows Search. Input “Common Motion Utility” and launch the application. Wait patiently for the application to startup.

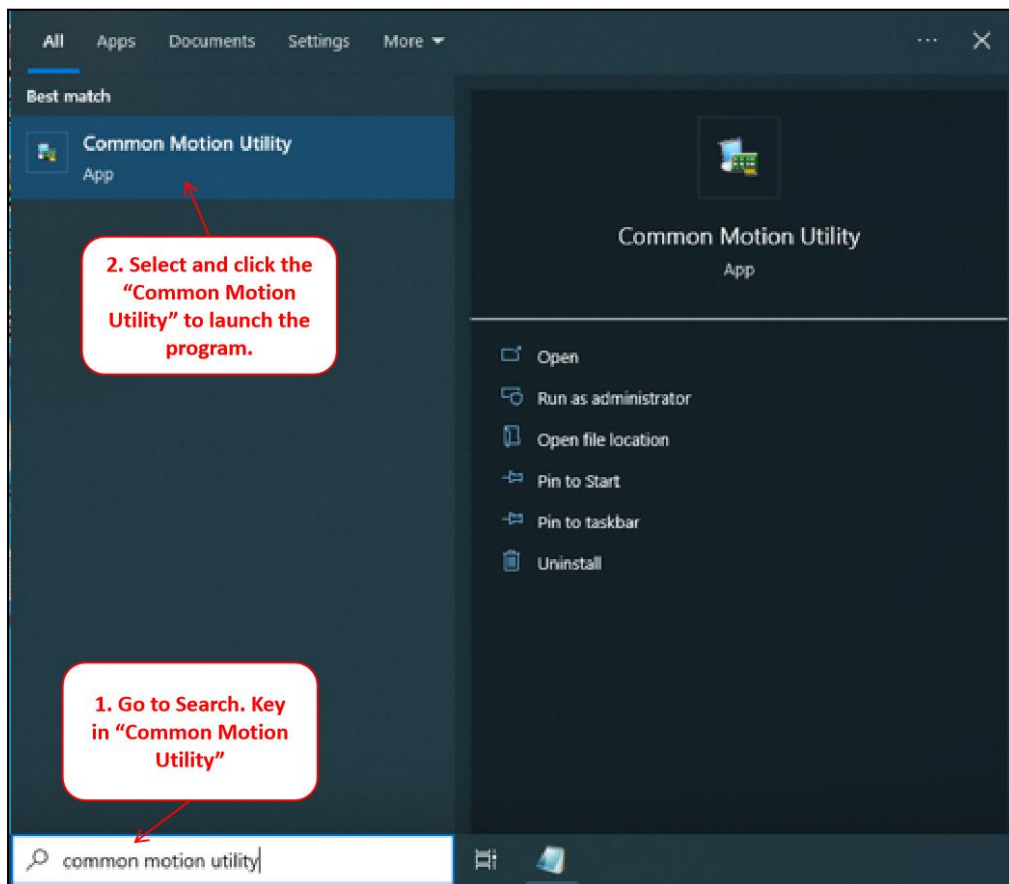


Figure 81: Launch Common Motion Utility Application

	Work Instruction	Doc. Revision	00
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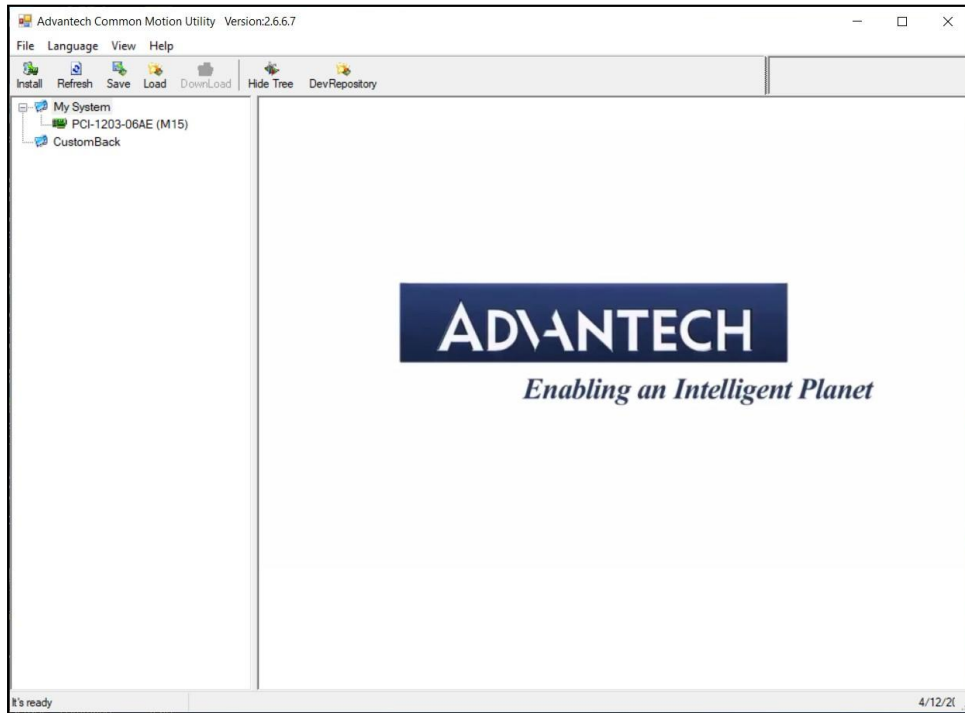


Figure 82: GUI for Common Motion Utility Startup

90. **Expand the list for PCI-1203-06AE (M15):** Double click the PCI-1203-06AE (M15) to expand its selection list. Under the Motion Ring node, there are 2 items as per below.
 - 90.1 0x001: MCDLT35BF: For X-axis home position and limit sensor adjustment
 - 90.2 0x002: MCDLT35BF: For Y-axis home position and limit sensor adjustment

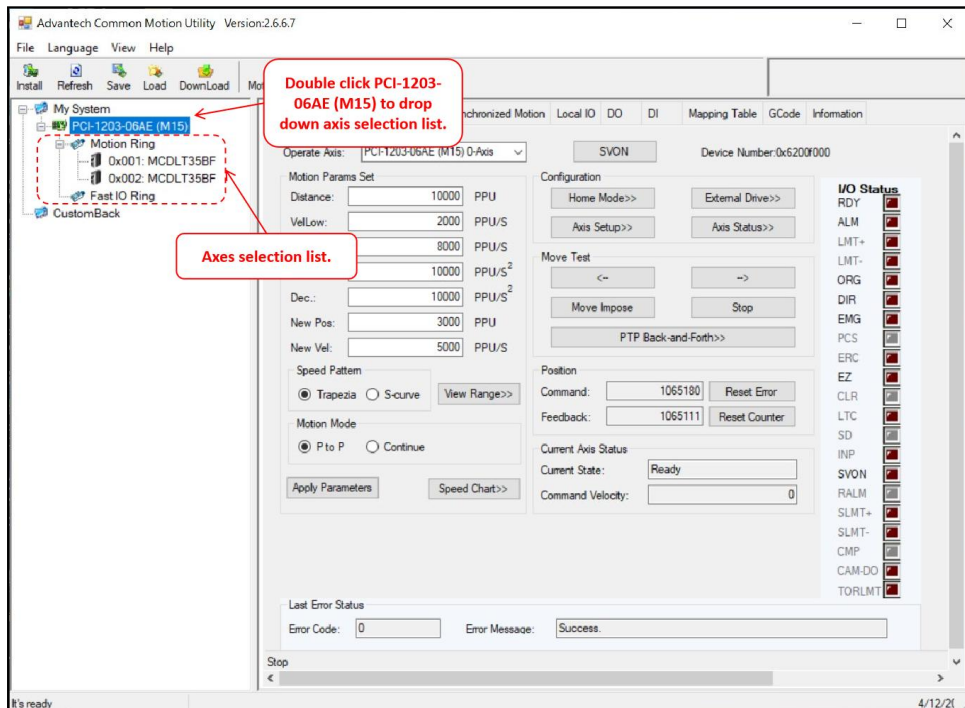


Figure 83: List Expansion for PCI-1203-06AE (M15)

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91. **Axis Selection:** Select the 0x002: MCDLT35BF to proceed home position adjustment for x-axis.

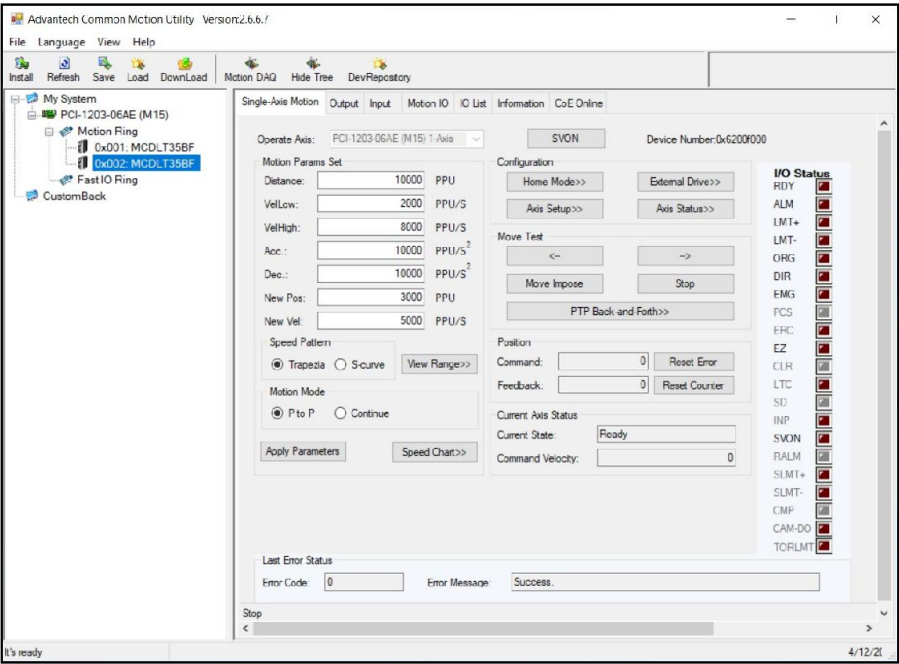


Figure 84: Axis Selection for Home Position Adjustment

92. **Ensure Limit - is Enabled at Hard Stop Position:** Upon axis selection, move the specific axis stage to hard stop position and ensure the limit - is in triggered mode. Else, adjust the limit sensor for signal to be triggered.

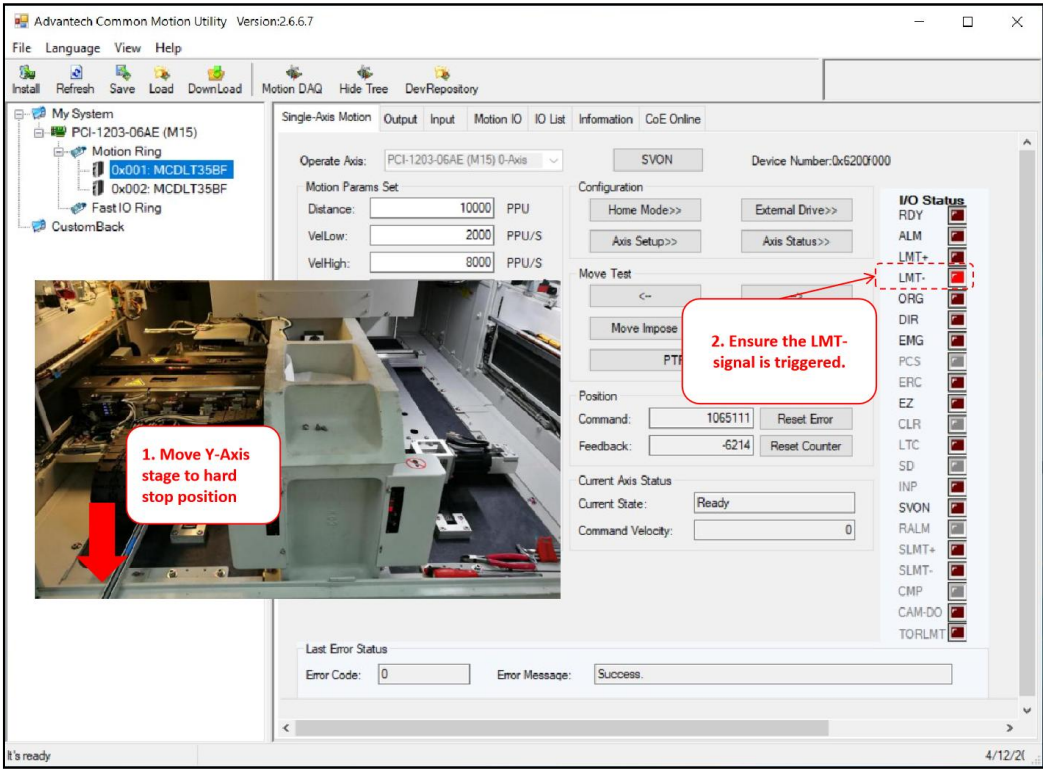


Figure 85: LMT- Signal Triggering Status

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93. **Y-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the Y-Axis stage away from hard stop position.

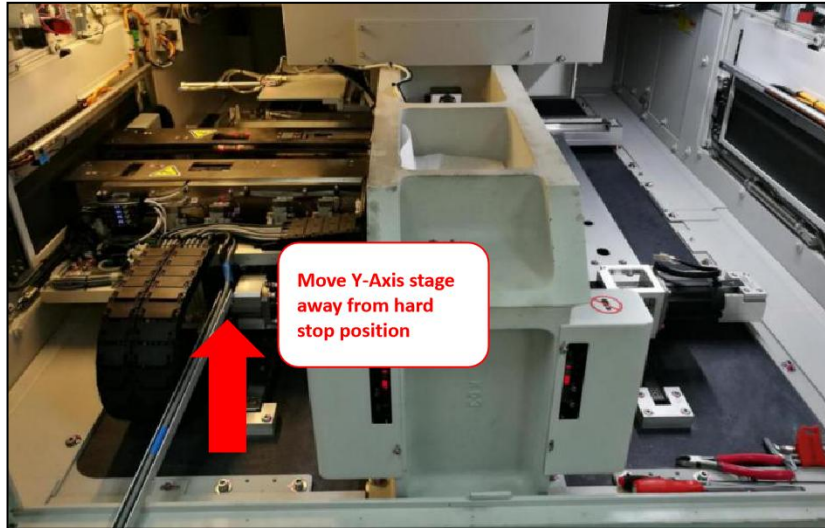


Figure 86: Move Y-Axis Stage Away From Hard Stop

94. **Perform Homing:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 2 for y-axis to go home position.

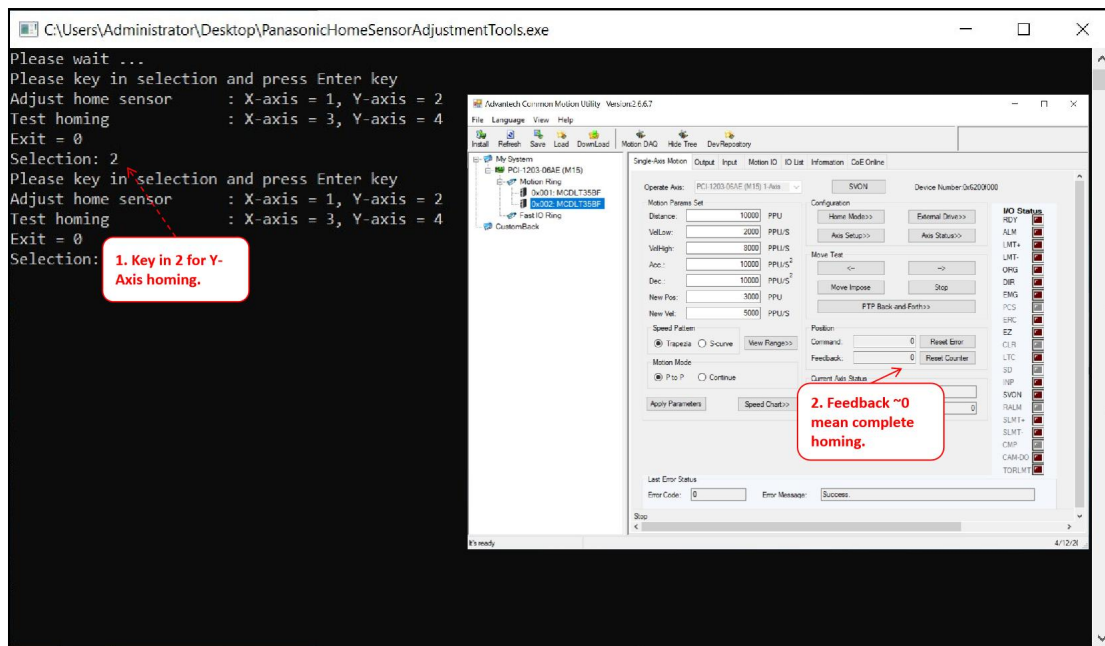


Figure 87: Perform Homing

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95. **Move the Y-Stage to Hard Stop Position:** Move the Y-stage to the hard stop position.

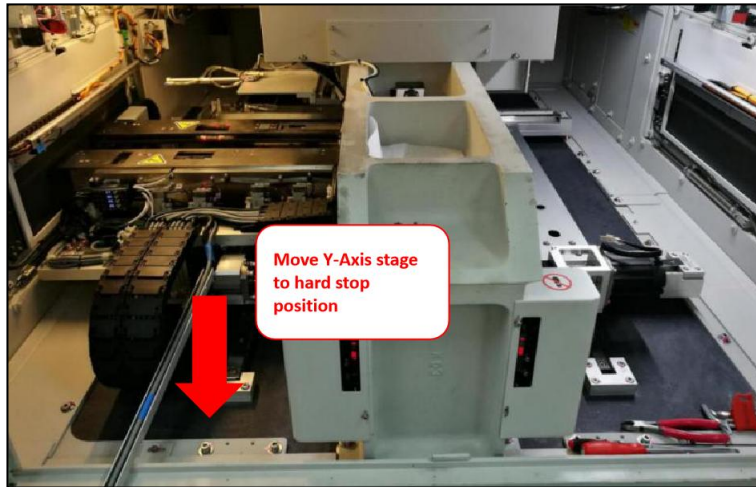


Figure 88: Move Y-Stage to Hard Stop Position

96. **Validate the Feedback Position is 4980-5180:** At the hard stop position, validate the feedback position. If it is not within acceptable range of 4980-5180, please proceed with Step 97. Else, proceed to Step 101.

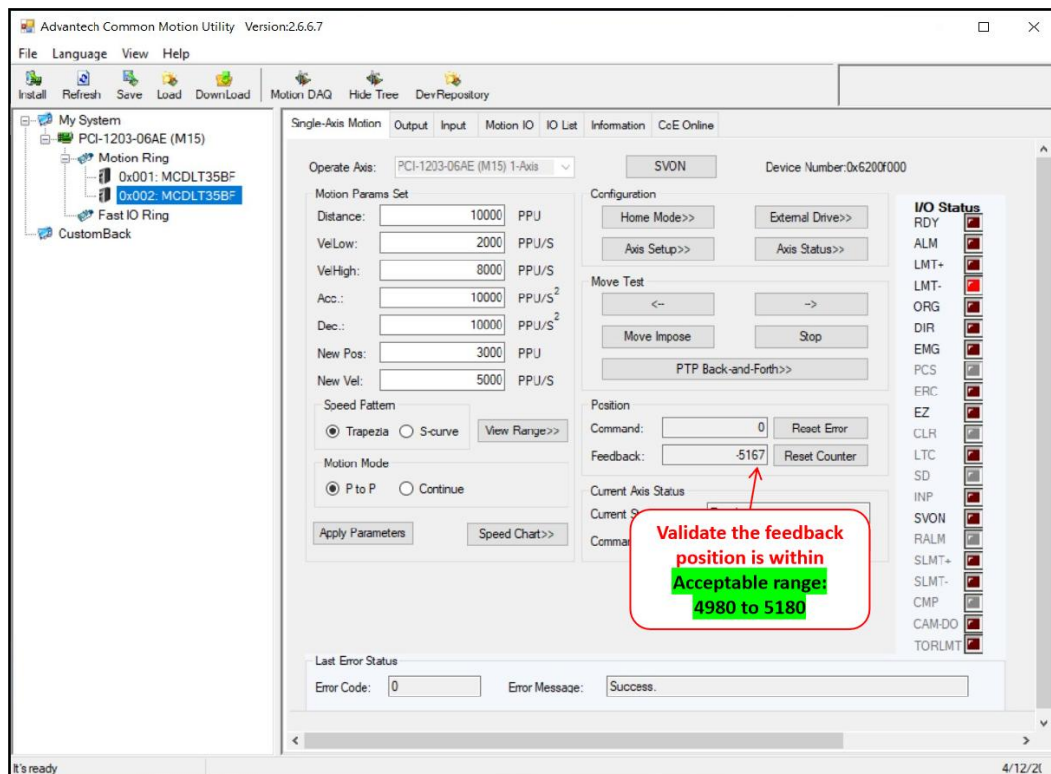


Figure 89: Feedback Position Validation

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97. **Loosen Spring Clamp:** Loosen all the screws for the spring clamp.

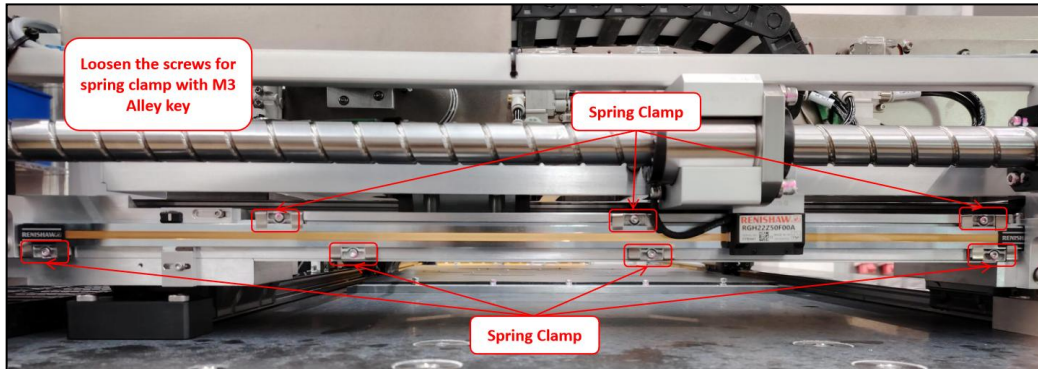


Figure 90: Quantic Scale Spring Clamps (Sample Image)

98. **Quantic Scale Positioning Adjustment:** Gently move the Quantic scale. While moving the Quantic scale, check the “Feedback Position” in Common Motion Utility and ensure the reading fall in the range 4980 to 5180.

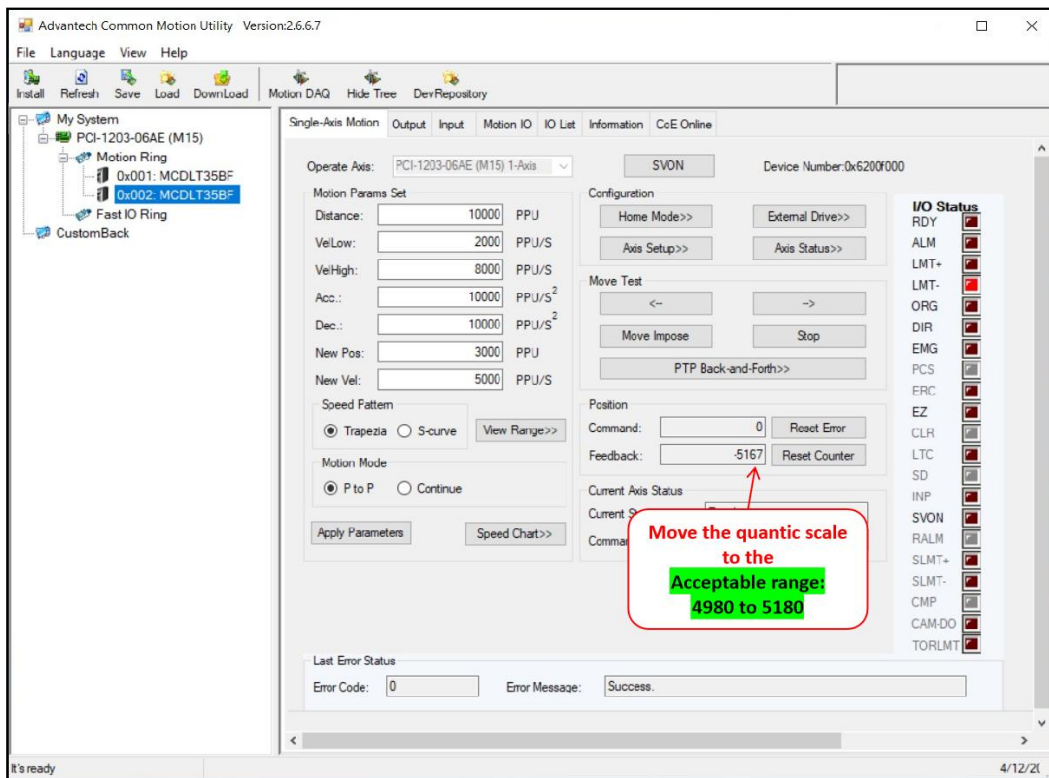


Figure 91: Quantic Scale Adjustment by Monitoring the Feedback Position

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99. **Tighten the Spring Clamps:** Upon completion of home position alignment, tighten all screw for the spring clamps.

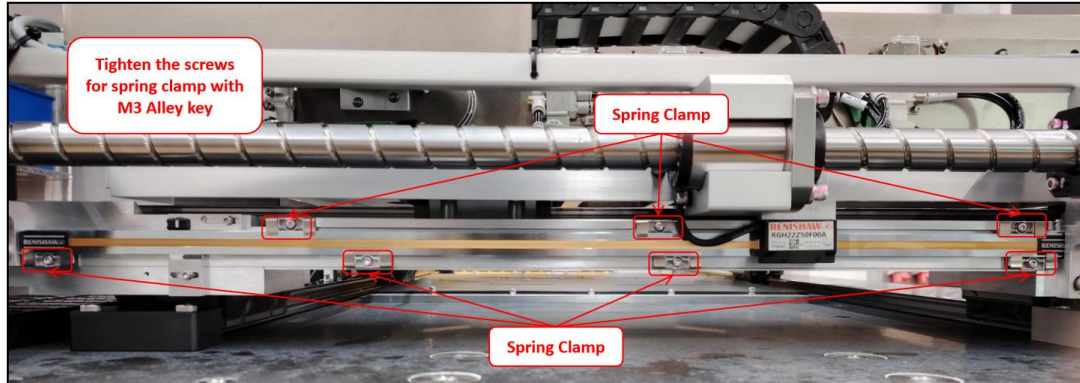


Figure 92: Y-Axis Quantic Scale Spring clamp (Sample Image)

100. **Home Sensor Positioning Offset:** Move the position of the home sensor to cross the engraved line mark at the top side of the Quantic scale.

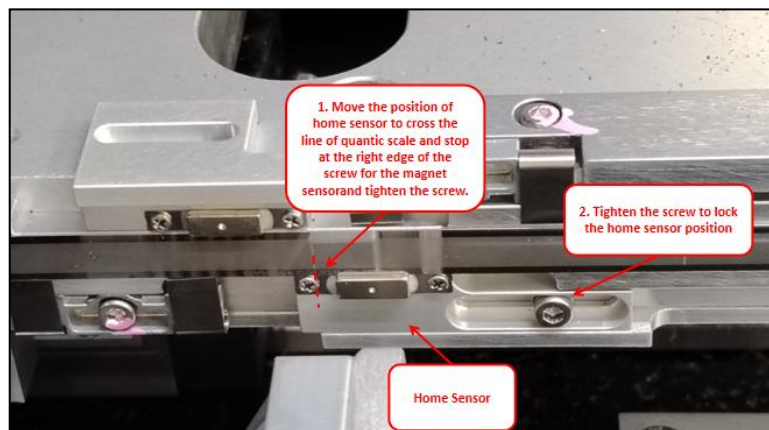


Figure 93: Y-Axis Home Sensor Offset

101. **Y-Axis Positioning (Away from Hard Stop):** Slightly and carefully move the Y-Axis to the middle of the stage.

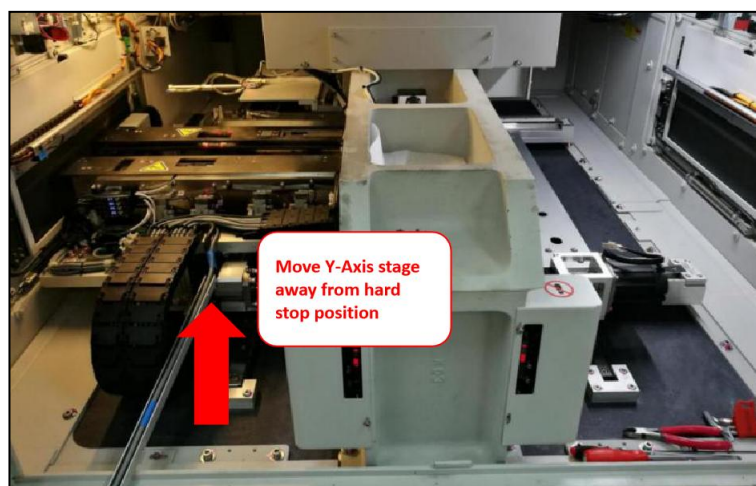


Figure 94: Move Y-Axis Stage Away From Hard Stop

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102. **Homing again:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 1 for x-axis to go home position.

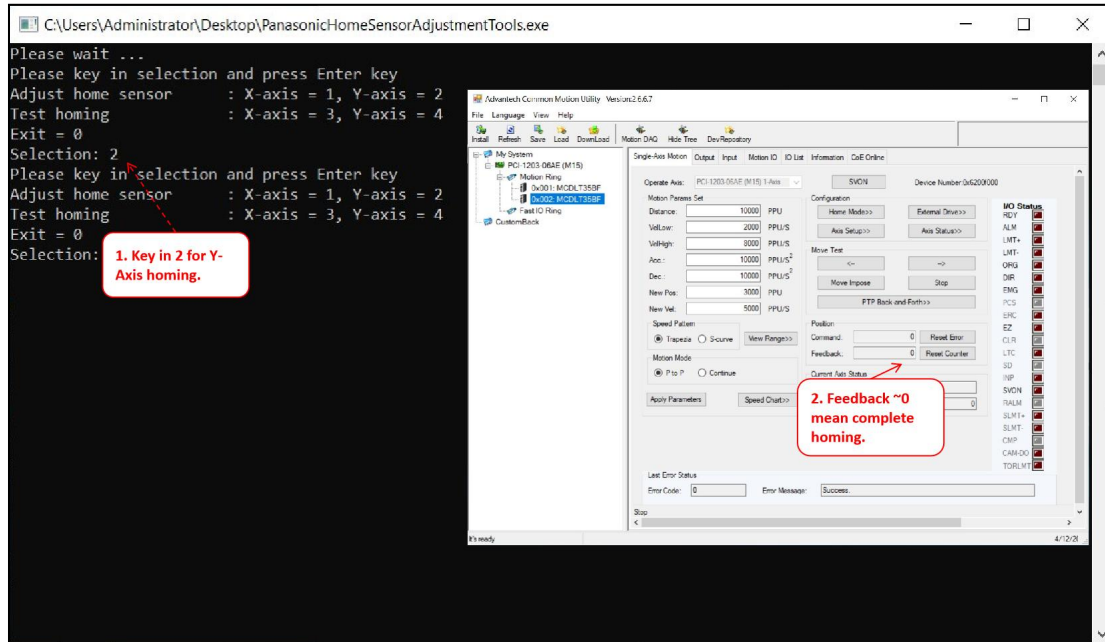


Figure 95: Perform Homing

103. **Move the Stage towards Hard Stop direction until it hit 1755 ± 1 :** Move the stage towards the hard stop direction and ensure the feedback position is within the range of 1754 to 1756.

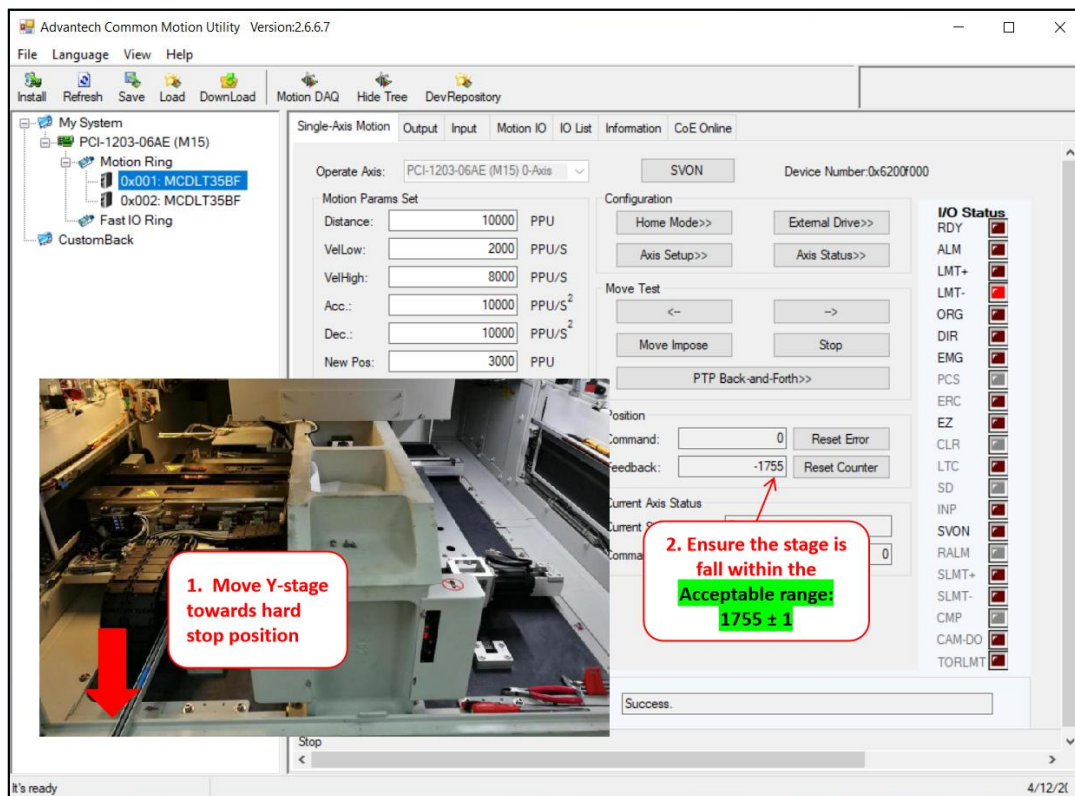


Figure 96: Move Y-Axis Stage to Feedback Position 1755 (Sample Image)

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104. **Move the limit sensor to most left position:** Positioning the limit sensor to the most left corner.

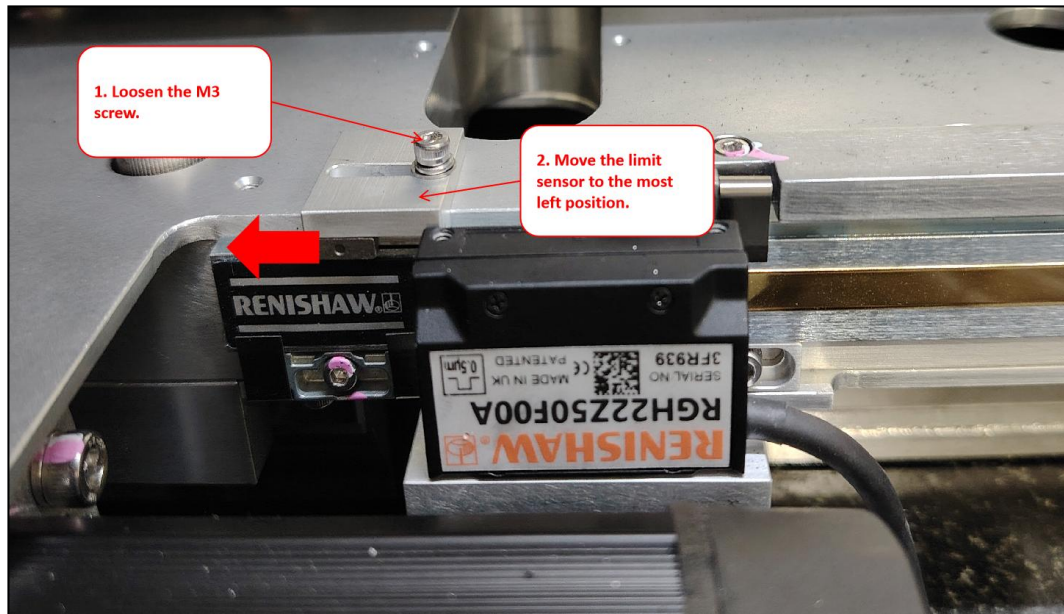


Figure 97: Move limit sensor to the Most Left Position (Sample Image)

105. **Adjust the limit sensor to right hand side until LMT- is triggered:** Slowly move the limit sensor towards the right hand side position until the LMT- signal is triggered.

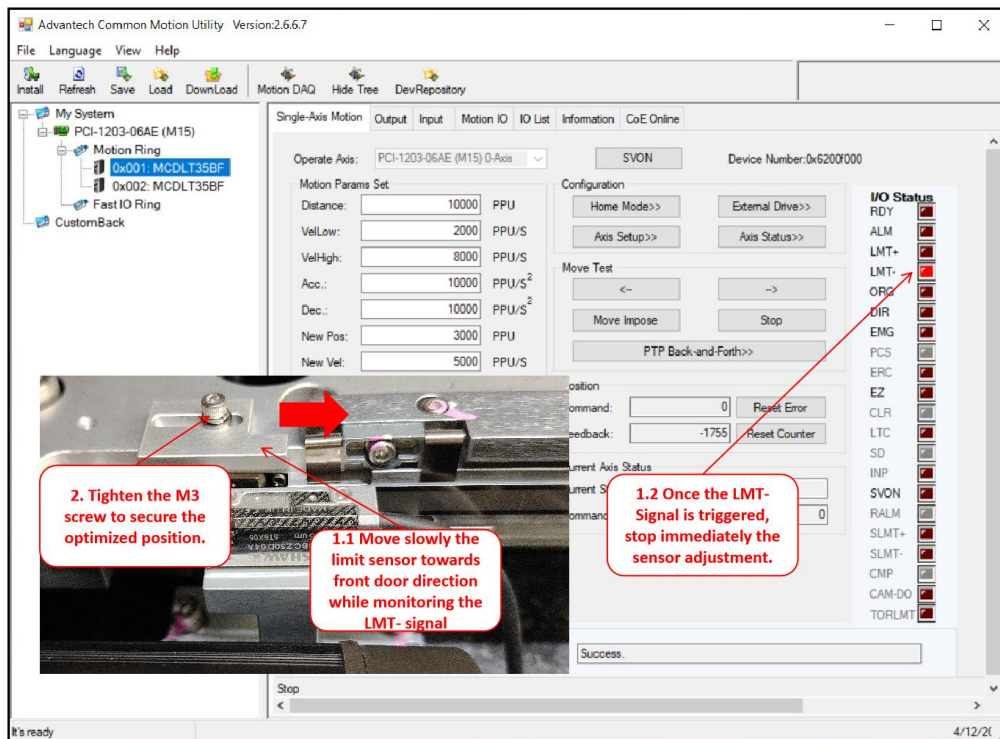


Figure 98: Positioning Limit Sensor to Triggered State (Sample Image)

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106. **Proceed to Test Homing for Final Verification:** Navigate to PanasonicHomeSensorAdjustment.exe program window. Key in selection 4 for y-axis to test homing result.

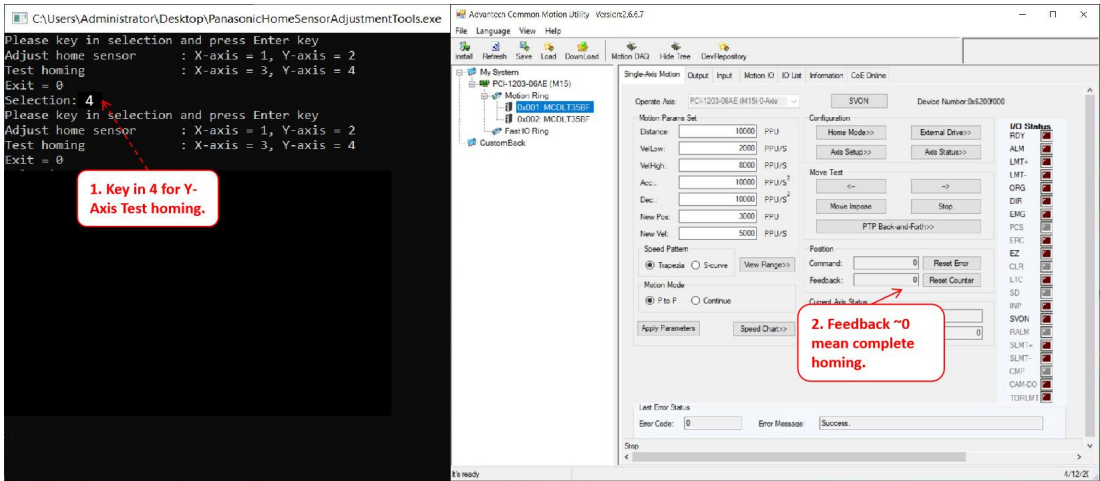


Figure 99: Perform Homing for Final Verification

107. **Re-connect the LAN cable labeled with Slice IO to the Atom PC:** Re-connect the LAN cable labeled with Slice IO to the Atom PC.



Figure 100: Reconnect the LAN Cable for Slice IO

108. **Go to Main Server and open v810:** Close the remote desktop connection and launch the v810 program.
109. **Panel Positioner Initialization:** Navigate to ‘Service > Subsystem Status & Control > Panel Positioner’. Click “Initialize”.

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110. **Verify the “Current Stage Position” for Y-Axis at hard stop position is within range of 3.08mm to 3.28mm:** Manually move the Y stage to the hard stop position and verify the home sensor reading is within 3.08mm to 3.28mm. If not within the range, route back to Step 78 to redo the adjustment all over again.

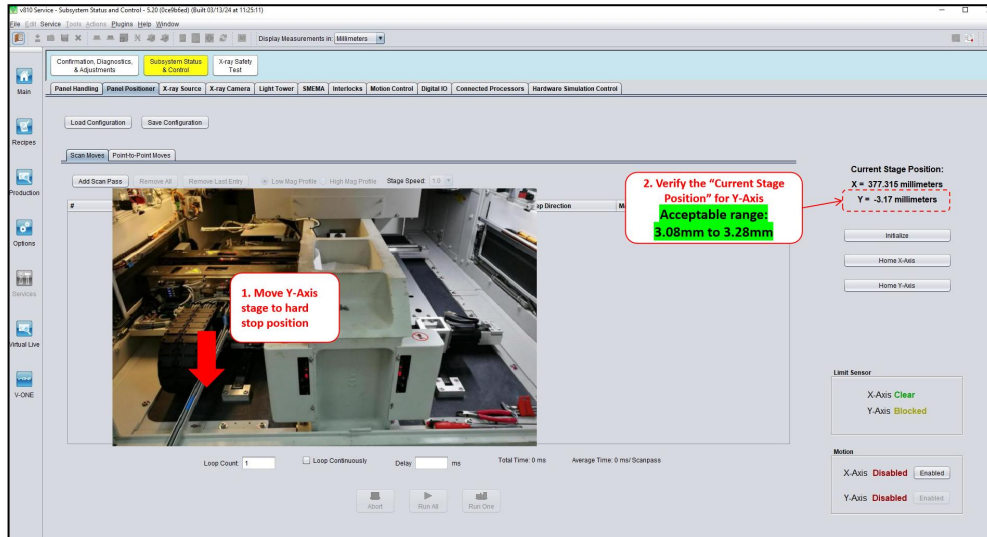


Figure 101: Current Stage Position for Y-Axis within range (Sample Image)

111. **Proceed to Validate Machine Performance:** Validation of machine is required before release the machine to production. Refer to “**Validation of Machine Performance After Alignment for Sensors**” procedure in last page.

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Validation of Machine Performance After Alignment for Sensors

112. **Tool Removal:** After completing the alignment process, ensure all tools, equipment, and debris are removed from the machine stage and surrounding area. Collect any tools used during the adjustment and confirmation procedures and return them to their designated storage locations.
113. **Access Door Switching:** Proceed to each access door of the machine and switch them back to production mode. Ensure that each access door is securely closed and locked according to standard operating procedures and verify that all safety interlocks associated with the access doors are engaged and functioning properly.
114. **Subsystem Initialization:** Power on the machine and run V810 software. Perform a login procedure to access the system interface. Navigate to the 'Services > Subsystem Status & Control > Panel Positioner' section within the software interface. Click on the "Initialize" button to reset the panel positioner subsystem to its default settings. Wait for the initialization process to complete before proceeding further.
115. **Long Scan Execution:** Execute five loops of long scan procedures using the machine's scanning capabilities. Ensure that the scan parameters are set according to the desired specifications and imaging requirements. Monitor the scan progress and observe the performance of the machine throughout the scanning process.
116. **CD&A Loop Execution:** Run one loop of CD&A (Comparison, Dosage, and Alignment) procedures to further validate the performance of the machine. Follow the standard operating procedures for CD&A loop execution, ensuring that all necessary parameters and settings are configured correctly.
117. **Monitoring and Verification of Machine Performance:** Monitor the execution of CD&A procedures and carefully observe the machine's performance, including accuracy, stability, and consistency in delivering results. Verify that the machine operates within acceptable tolerances and meets the required performance standards.